0001tumour

| | |
|---|---|
| Report date | Apr 5, 2021 8:33:06 PM |

Contents



# 1. Global Definitions

| | |
|---|---|
| Date | Apr 5, 2021 8:32:01 PM |

Global settings

| | |
|---|---|
| Name | 0001tumour.mph |
| Path | C:\Users\sruth\Desktop\Final Year Project\COMSOL\Final Working RFA\0001tumour.mph |
| Version | COMSOL Multiphysics 5.6 (Build: 280) |
| Unit system | SI |

Used products

| |
|---|
| COMSOL Multiphysics |
| CAD Import Module |
| Heat Transfer Module |

Computer information

| | |
|---|---|
| CPU | Intel64 Family 6 Model 142 Stepping 9, 2 cores |
| Operating system | Windows 10 |

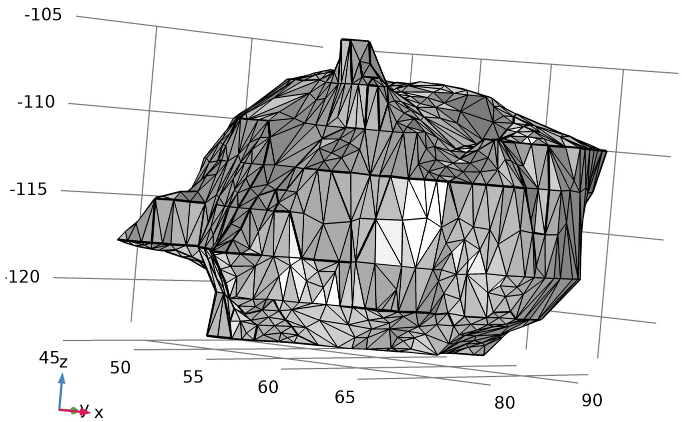
## 1.1. Parameters

Parameters 1

| Name | Expression | Value | Description |
|---|---|---|---|
| rho_b | 1000[kg/m^3] | 1000 kg/m³ | Density, blood |
| c_b | 4180[J/(kg*K)] | 4180 J/(kg·K) | Heat capacity, blood |
| omega_b | 6.4e-3[1/s] | 0.0064 1/s | Blood perfusion rate |
| T_b | 37[degC] | 310.15 K | Arterial blood temp |
| T0 | 37[degC] | 310.15 K | Initial and boundary temp |
| V0 | 22[V] | 22 V | Electric voltage |
| xc_v | 26[mm] | 0.026 m | Vessel cylinder center x-coordinate |
| a_time | 10[min] | 600 s | Ablation time |

## 1.2. Mesh Parts

### 1.2.1. Mesh Part 1




*Mesh Part 1*

Mesh statistics

| Description | Value |
|---|---|
| Minimum element quality | 1.582E-6 |
| Average element quality | 0.5934 |
| Triangle | 2308 |
| Edge element | 497 |
| Vertex element | 143 |

**Import 1 (imp1)**

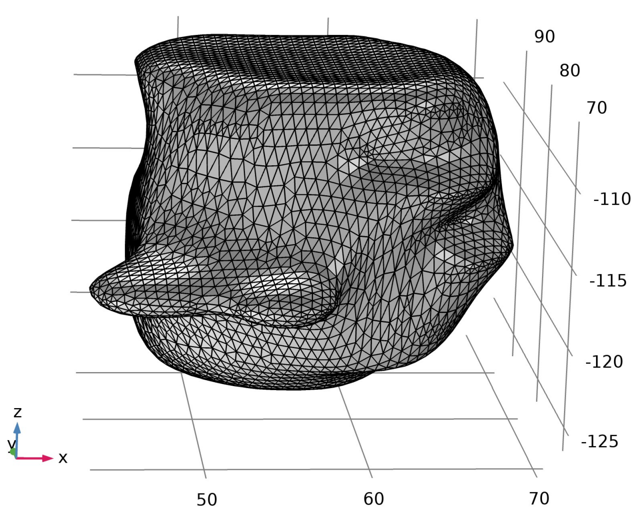
Settings

| Description | Value |
|---|---|
| Source | STL file |
| Filename | C:\Users\sruth\Desktop\Final Year Project\MIMICS\MIMICS new stl\0001.stl |

Settings

| Name | SOLID section in file |
|---|---|
| Boundaries Import 1 | |

**1.2.2. Mesh Part 2**




*Mesh Part 2*

Mesh statistics

| Description | Value |
|---|---|
| | |

| | |
|---|---|
| Minimum element quality | 0.142 |
| Average element quality | 0.8175 |
| Triangle | 8032 |

##### Import 1 (imp1)

Settings

| Description | Value |
|---|---|
| Source | STL file |
| Filename | C:\Users\sruth\Desktop\Final Year Project\3D slicer\3D slicer segmentation files\0001\Segmentation_Tumor 001.stl |

Settings

| Name | SOLID section in file |
|---|---|
| Boundaries Import 1 | |

## 2. Component 1

| | |
|---|---|
| Date | Mar 20, 2021 1:04:37 PM |

Settings

| Description | Value |
|---|---|
| Unit system | Same as global system (SI) |
| Geometry shape function | Automatic |

Spatial frame coordinates

| First | Second | Third |
|---|---|---|
| x | y | z |

Material frame coordinates

| First | Second | Third |
|---|---|---|
| X | Y | Z |

Geometry frame coordinates

| First | Second | Third |
|---|---|---|
| Xg | Yg | Zg |

Mesh frame coordinates

| First | Second | Third |
|---|---|---|
| Xm | Ym | Zm |

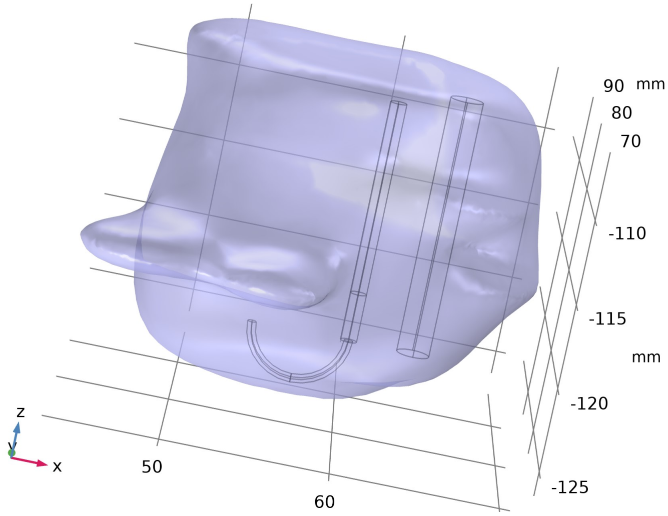
### 2.1. Definitions

#### 2.1.1. Selections

##### Exterior Boundaries

| Selection type |
|---|
| Explicit |

| Selection |
|---|
| Boundary 1 |

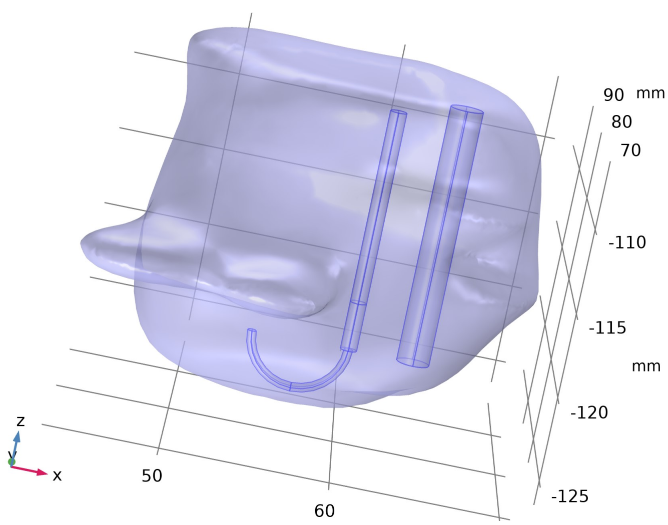



*Exterior Boundaries*

**Liver Tissue**

| Selection type |
| --- |
| Explicit |

| Selection |
| --- |
| Domain 1 |

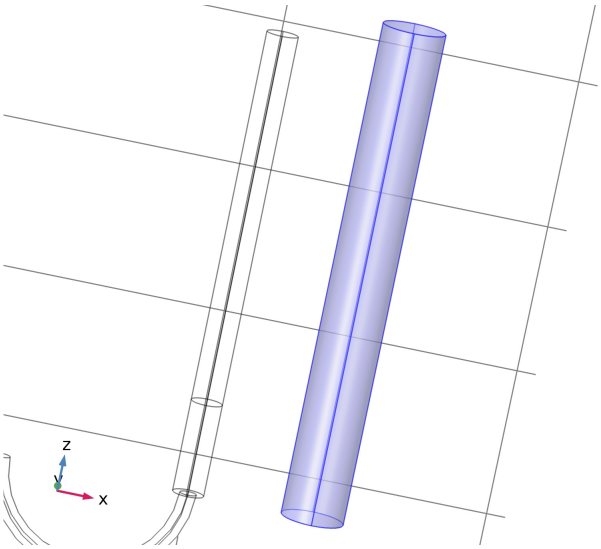



*Liver Tissue*

**Blood Vessel**

| Selection type |
| --- |
| Explicit |

| Selection |
| --- |
| Domain 5 |

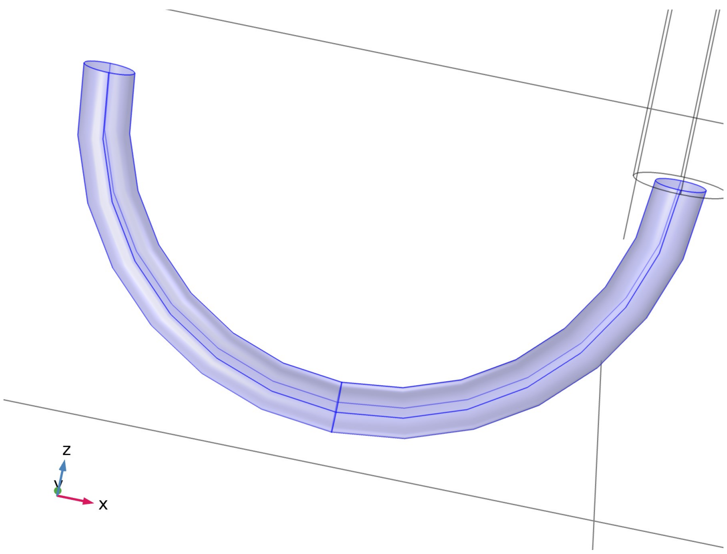



*Blood Vessel*

**Electrodes**

| **Selection type** |
|---|
| Explicit |

| **Selection** |
|---|
| Domain 2 |

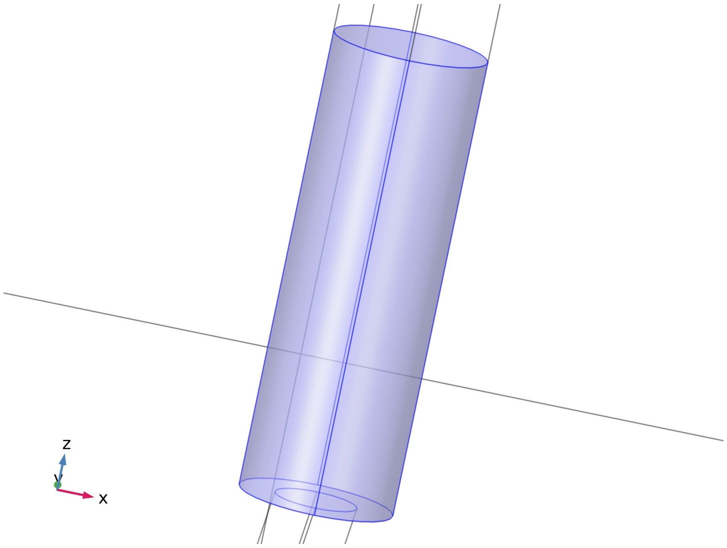



*Electrodes*

**Trocar Tip**

| **Selection type** |
|---|
| Explicit |

| **Selection** |
|---|
| Domain 3 |

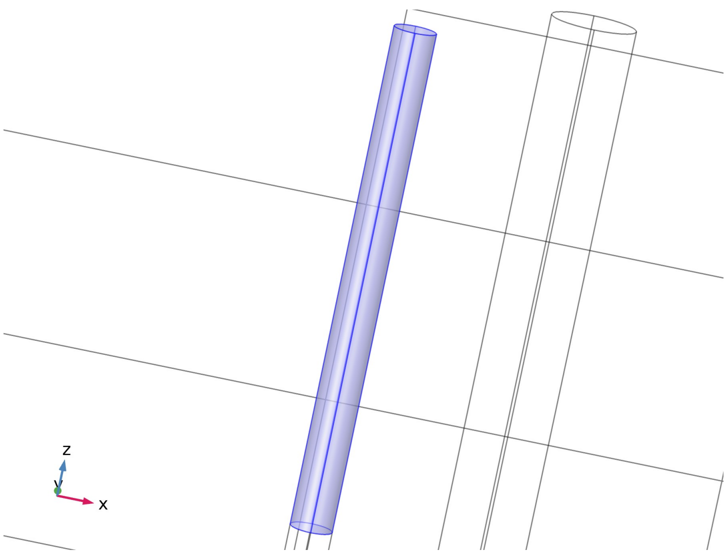



*Trocar Tip*

**Trocar Base**

| Selection type |
| --- |
| Explicit |

| Selection |
| --- |
| Domain 4 |




*Trocar Base*

**Trocar**

| Selection type |
| --- |
| Union |

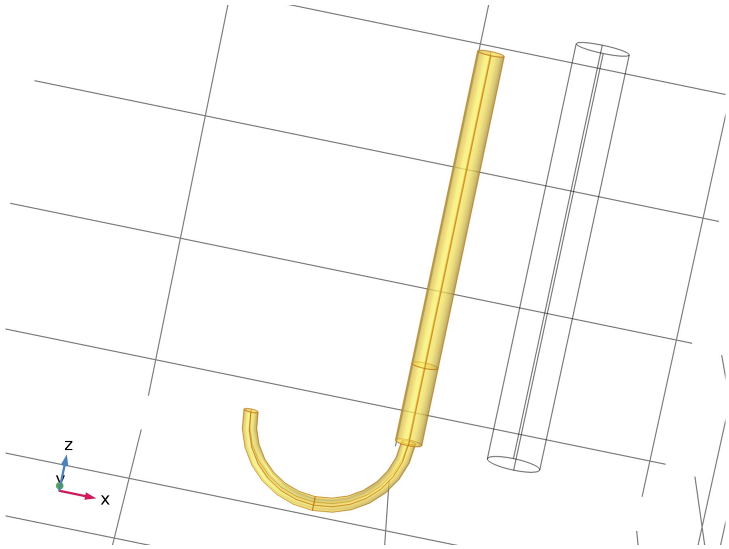
| Selection |
| --- |
| Domains 2–4 |

Geometric entity level

| Description | Value |
| --- | --- |
| Level | Domain |

Input entities

| Description | Value |
| --- | --- |
| Selections to add | {Electrodes, Trocar Tip, Trocar Base} |




*Trocar*

**Tissue and Trocar**

| Selection type |
|---|
| Union |

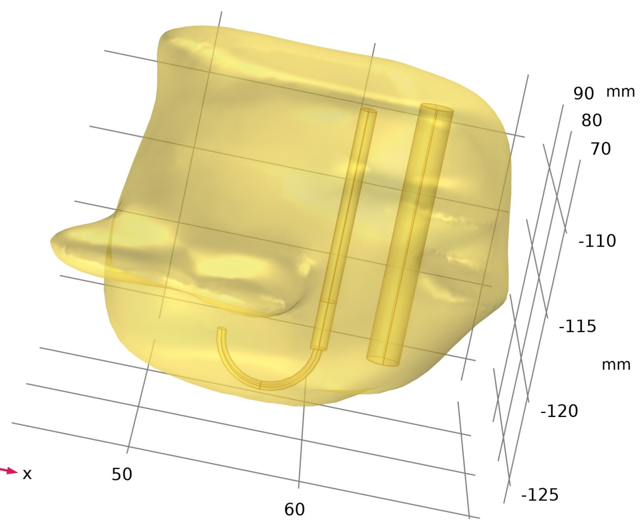
| Selection |
|---|
| Domains 1–4 |

Geometric entity level

| Description | Value |
|---|---|
| Level | Domain |

Input entities

| Description | Value |
|---|---|
| Selections to add | {Liver Tissue, Trocar} |




*Tissue and Trocar*

**Tissue and Trocar, Exterior Boundaries**

| Selection type |
|---|
| Adjacent |

| Selection |
|---|
| Boundaries 1, 23–28 |

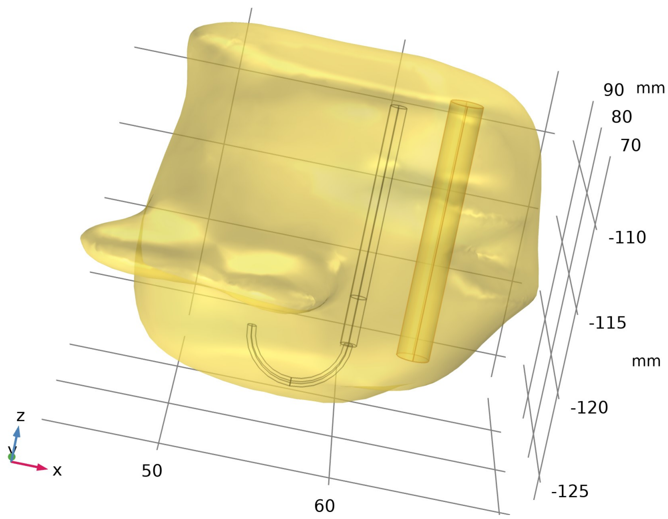
Input entities

| Description | Value |
|---|---|
| | |

| | |
|---|---|
| Geometric entity level | Domain |
| Input selections | Tissue and Trocar |

Output entities

| Description | Value |
|---|---|
| Geometric entity level | Adjacent boundaries |
| Exterior boundaries | On |
| Interior boundaries | Off |




*Tissue and Trocar, Exterior Boundaries*

**Trocar Tip and Electrodes, Exterior boundaries**

| Selection type |
|---|
| Adjacent |

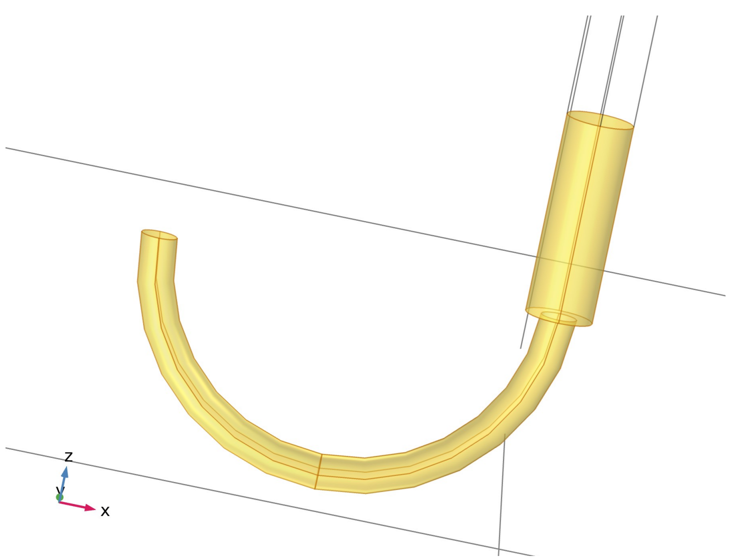
| Selection |
|---|
| Boundaries 2–13, 16, 19, 21 |

Input entities

| Description | Value |
|---|---|
| Geometric entity level | Domain |
| Input selections | {Electrodes, Trocar Tip} |

Output entities

| Description | Value |
|---|---|
| Geometric entity level | Adjacent boundaries |
| Exterior boundaries | On |
| Interior boundaries | Off |



*Trocar Tip and Electrodes, Exterior boundaries*

#### 2.1.2. Probes

##### Domain Point Probe 1

| | |
|---|---|
| Probe type | Domain point probe |

#### 2.1.3. Coordinate Systems

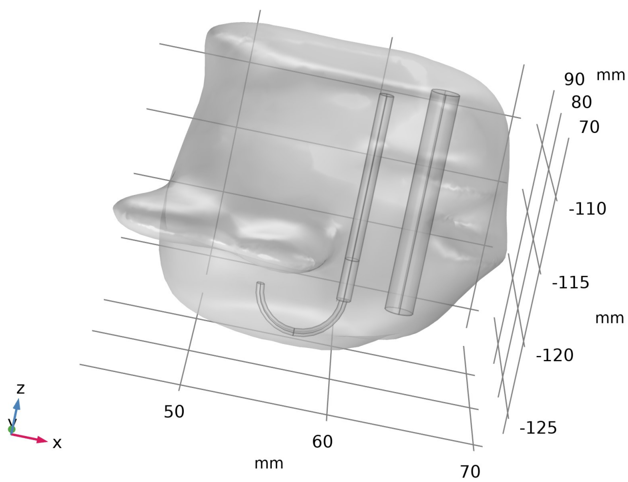
##### Boundary System 1

| | |
|---|---|
| Coordinate system type | Boundary system |
| Tag | sys1 |

Coordinate names

| First | Second | Third |
|---|---|---|
| t1 | t2 | n |

### 2.2. Geometry 1




*Geometry 1*

Units

| | |
|---|---|
| Length unit | mm |
| Angular unit | deg |

Geometry statistics

| Description | Value |
|---|---|
| Space dimension | 3 |
| Number of domains | 5 |
| Number of boundaries | 28 |
| Number of edges | 52 |
| Number of vertices | 32 |

#### 2.2.1. Cylinder 1 (cyl1)

Position

| Description | Value |
|---|---|
| Position | {60, 85, -123} |

Axis

| Description | Value |
|---|---|
| Axis type | z - axis |
| Layers on side | Off |
| Layers on bottom | On |

Axis

| Layer name | Thickness (mm) |
|---|---|
| Layer 1 | 3 |

Size and shape

| Description | Value |
|---|---|

| | |
|---|---|
| Radius | 0.5 |
| Height | 15 |

#### 2.2.2. Torus 1 (tor1)

Position

| Description | Value |
|---|---|
| Position | {57, 85, -123} |

Rotation angle

| Description | Value |
|---|---|
| Rotation | -90 |

Axis

| Description | Value |
|---|---|
| Axis type | y - axis |

Size and shape

| Description | Value |
|---|---|
| Major radius | 3 |
| Minor radius | 0.2667 |
| Revolution angle | 180 |

#### 2.2.3. Cylinder 2 (cyl2)

Position

| Description | Value |
|---|---|
| Position | {64, 85, -123} |

Axis

| Description | Value |
|---|---|
| Axis type | z - axis |

Size and shape

| Description | Value |
|---|---|
| Radius | 1 |
| Height | 16 |

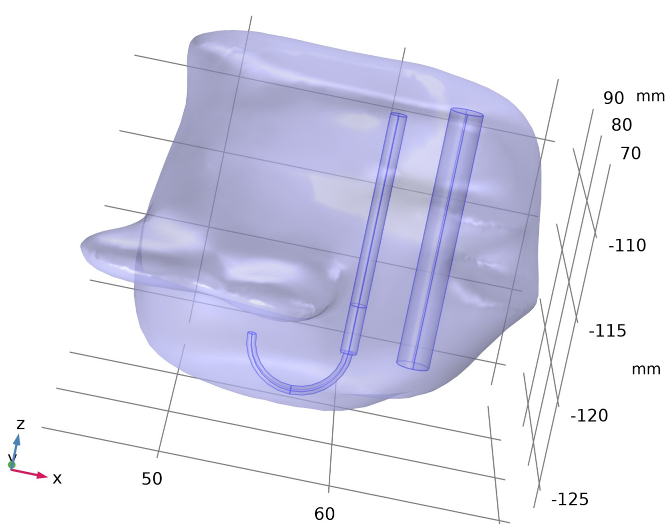
#### 2.2.4. Import 1 (imp1)

Settings

| Description | Value |
|---|---|
| Source | Mesh or 3D printing file (STL, 3MF, PLY) |
| Mesh | Mesh Part 2 |

### 2.3. Materials

#### 2.3.1. Liver (human)




*Liver (human)*

Selection

| | |
|---|---|
| Geometric entity level | Domain |
| Name | Liver Tissue |
| Selection | Named sel2: Geometry geom1: Dimension 3: Domain 1 |

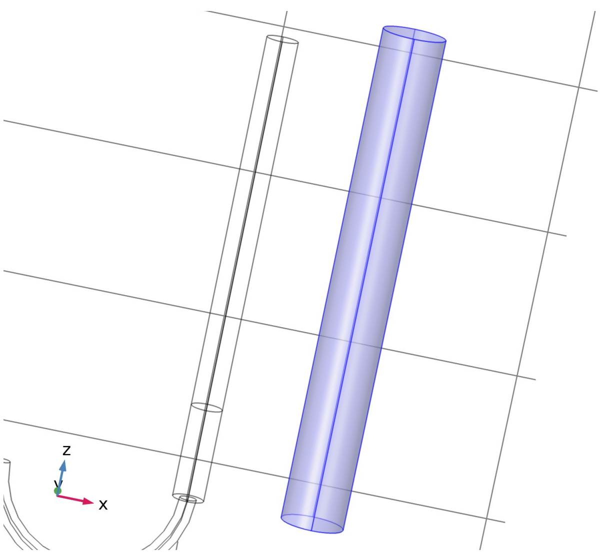
Material parameters

| Name | Value | Unit |
|---|---|---|
| Heat capacity at constant pressure | 3540[J/(kg*K)] | J/(kg·K) |
| Density | 1079[kg/m^3] | kg/m³ |
| Thermal conductivity | 0.52[W/(m*K)] | W/(m·K) |
| Frequency factor | 7.39e39 | 1/s |
| Activation energy | 2.577e5 | J/mol |
| Electrical conductivity | 0.333[S/m] | S/m |
| Relative permittivity | 1 | 1 |

Basic

| Description | Value |
|---|---|
| Heat capacity at constant pressure | 3540[J/(kg*K)] |
| heatcapacity_symmetry | 0 |
| Density | 1079[kg/m^3] |
| density_symmetry | 0 |
| Thermal conductivity | {{0.52[W/(m*K)], 0, 0}, {0, 0.52[W/(m*K)], 0}, {0, 0, 0.52[W/(m*K)]}} |
| thermalconductivity_symmetry | 3 |
| Frequency factor | 7.39e39 |
| frequencyfactor_symmetry | 0 |
| Activation energy | 2.577e5 |
| activationenergy_symmetry | 0 |
| Electrical conductivity | {{0.333[S/m], 0, 0}, {0, 0.333[S/m], 0}, {0, 0, 0.333[S/m]}} |
| electricconductivity_symmetry | 0 |
| Relative permittivity | {{1, 0, 0}, {0, 1, 0}, {0, 0, 1}} |
| relpermittivity_symmetry | 0 |

**2.3.2. Blood**




*Blood*

Selection

| | |
|---|---|
| Geometric entity level | Domain |
| Name | Blood Vessel |
| Selection | Named sel3: Geometry geom1: Dimension 3: Domain 5 |

Material parameters

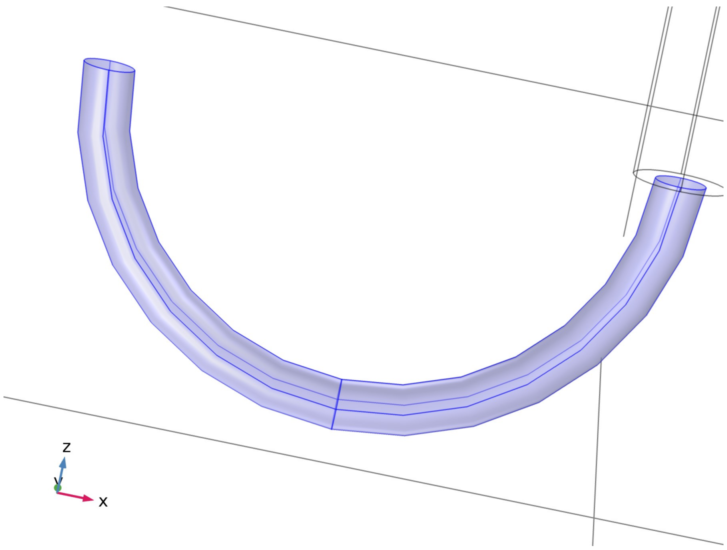
| Name | Value | Unit |
|---|---|---|
| Electrical conductivity | 0.667[S/m] | S/m |
| Relative permittivity | 1 | 1 |

Basic

| Description | Value |
|---|---|
| Electrical conductivity | {{0.667[S/m], 0, 0}, {0, 0.667[S/m], 0}, {0, 0, 0.667[S/m]}} |
| electricconductivity_symmetry | 0 |
| | |

| | |
|---|---|
| Relative permittivity | {{1, 0, 0}, {0, 1, 0}, {0, 0, 1}} |
| relpermittivity_symmetry | 0 |
| Thermal conductivity | {{0.543[W/(m*K)], 0, 0}, {0, 0.543[W/(m*K)], 0}, {0, 0, 0.543[W/(m*K)]}} |
| thermalconductivity_symmetry | 0 |
| Heat capacity at constant pressure | c_b |
| heatcapacity_symmetry | 0 |
| Density | rho_b |
| density_symmetry | 0 |

**2.3.3. Electrodes**




*Electrodes*

Selection

| | |
|---|---|
| Geometric entity level | Domain |
| Name | Electrodes |
| Selection | Named sel4: Geometry geom1: Dimension 3: Domain 2 |

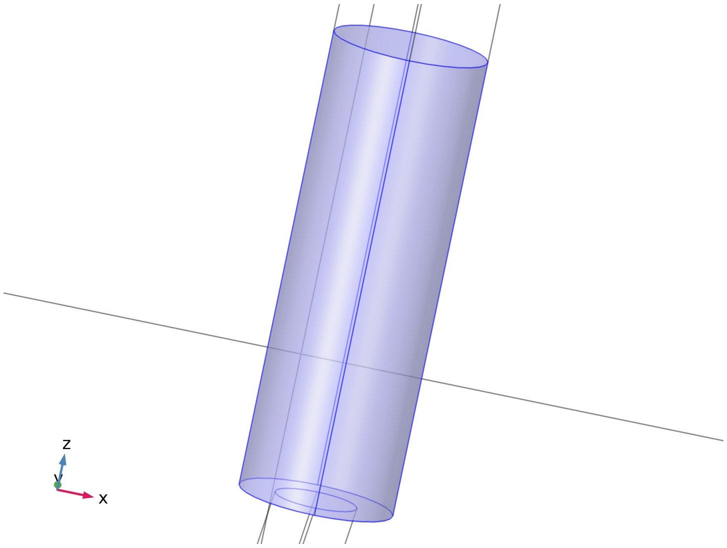
Material parameters

| Name | Value | Unit |
|---|---|---|
| Electrical conductivity | 1e8[S/m] | S/m |
| Relative permittivity | 1 | 1 |
| Density | 6450[kg/m^3] | kg/m³ |
| Heat capacity at constant pressure | 840[J/(kg*K)] | J/(kg·K) |
| Thermal conductivity | 18[W/(m*K)] | W/(m·K) |

Basic

| Description | Value |
|---|---|
| Electrical conductivity | {{1e8[S/m], 0, 0}, {0, 1e8[S/m], 0}, {0, 0, 1e8[S/m]}} |
| electricconductivity_symmetry | 0 |
| Relative permittivity | {{1, 0, 0}, {0, 1, 0}, {0, 0, 1}} |
| relpermittivity_symmetry | 0 |
| Density | 6450[kg/m^3] |
| density_symmetry | 0 |
| Heat capacity at constant pressure | 840[J/(kg*K)] |
| heatcapacity_symmetry | 0 |
| Thermal conductivity | {{18[W/(m*K)], 0, 0}, {0, 18[W/(m*K)], 0}, {0, 0, 18[W/(m*K)]}} |
| thermalconductivity_symmetry | 0 |

**2.3.4. Trocar Tip**




*Trocar Tip*

Selection

| | |
|---|---|
| Geometric entity level | Domain |
| Name | Trocar Tip |
| Selection | Named sel5: Geometry geom1: Dimension 3: Domain 3 |

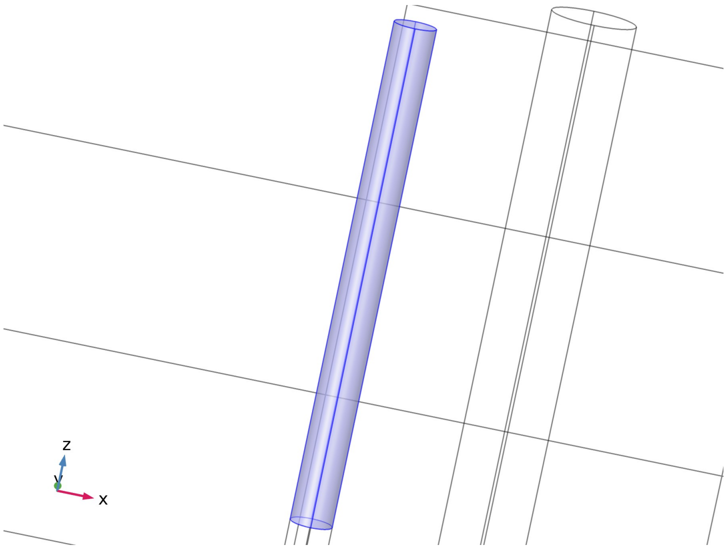
Material parameters

| Name | Value | Unit |
|---|---|---|
| Electrical conductivity | 4e6[S/m] | S/m |
| Relative permittivity | 1 | 1 |
| Density | 21500[kg/m^3] | kg/m³ |
| Heat capacity at constant pressure | 132[J/(kg*K)] | J/(kg·K) |
| Thermal conductivity | 71[W/(m*K)] | W/(m·K) |

Basic

| Description | Value |
|---|---|
| Electrical conductivity | {{4e6[S/m], 0, 0}, {0, 4e6[S/m], 0}, {0, 0, 4e6[S/m]}} |
| electricconductivity_symmetry | 0 |
| Relative permittivity | {{1, 0, 0}, {0, 1, 0}, {0, 0, 1}} |
| relpermittivity_symmetry | 0 |
| Density | 21500[kg/m^3] |
| density_symmetry | 0 |
| Heat capacity at constant pressure | 132[J/(kg*K)] |
| heatcapacity_symmetry | 0 |
| Thermal conductivity | {{71[W/(m*K)], 0, 0}, {0, 71[W/(m*K)], 0}, {0, 0, 71[W/(m*K)]}} |
| thermalconductivity_symmetry | 0 |

**2.3.5. Trocar Base**




*Trocar Base*

Selection

| | |
|---|---|
| Geometric entity level | Domain |
| Name | Trocar Base |
| Selection | Named sel6: Geometry geom1: Dimension 3: Domain 4 |

Material parameters

| Name | Value | Unit |
|---|---|---|
| Electrical conductivity | 1e-5[S/m] | S/m |
| Relative permittivity | 1 | 1 |
| Density | 70[kg/m^3] | kg/m³ |
| Heat capacity at constant pressure | 1045[J/(kg*K)] | J/(kg·K) |
| Thermal conductivity | 0.026[W/(m*K)] | W/(m·K) |

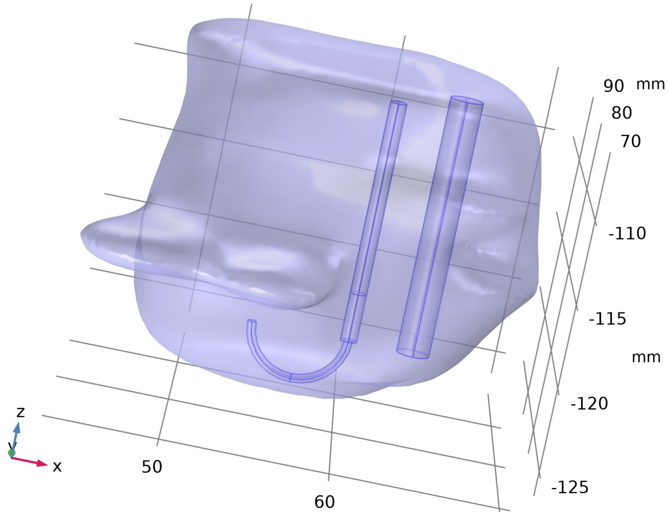
Basic

| Description | Value |
|---|---|
| Electrical conductivity | {{1e-5[S/m], 0, 0}, {0, 1e-5[S/m], 0}, {0, 0, 1e-5[S/m]}} |
| electricconductivity_symmetry | 0 |
| Relative permittivity | {{1, 0, 0}, {0, 1, 0}, {0, 0, 1}} |
| relpermittivity_symmetry | 0 |
| Density | 70[kg/m^3] |
| density_symmetry | 0 |
| Heat capacity at constant pressure | 1045[J/(kg*K)] |
| heatcapacity_symmetry | 0 |
| Thermal conductivity | {{0.026[W/(m*K)], 0, 0}, {0, 0.026[W/(m*K)], 0}, {0, 0, 0.026[W/(m*K)]}} |
| thermalconductivity_symmetry | 0 |

### 2.4. Electric Currents

Used products

| |
|---|
| COMSOL Multiphysics |




*Electric Currents*

Selection

| | |
|---|---|
| Geometric entity level | Domain |
| Selection | Geometry geom1: Dimension 3: All domains |

Equations

$$\nabla \cdot \mathbf{J} = Q_{j,v}$$

$$\mathbf{J} = \sigma \mathbf{E} + \mathbf{J}_e$$

$$\mathbf{E} = -\nabla V$$

### 2.4.1. Interface Settings

#### Discretization

Settings

| Description | Value |
|---|---|
| Electric potential | Linear |

#### Manual Terminal Sweep Settings

Settings

| Description | Value |
|---|---|
| Use manual terminal sweep | Off |
| Reference impedance | 50[ohm] |

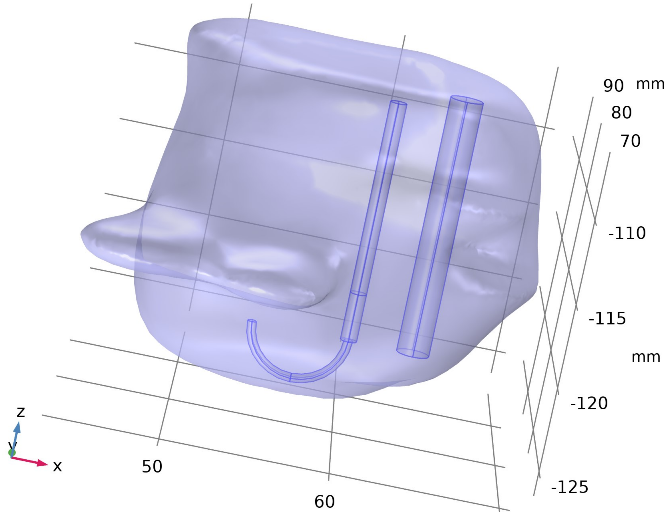
### 2.4.2. Variables

| Name | Expression | Unit | Description | Selection |
|---|---|---|---|---|
| ec.d | 1 | 1 | Contribution | Domains 1–5 |
| ec.I_sXX | (spatial.invF11*(spatial.invF11*ec.I_sxx+spatial.invF21*ec.I_syx+spatial.invF31*ec.I_szx)+spatial.invF21*(spatial.invF11*ec.I_sxy+spatial.invF21*ec.I_syy+spatial.invF31*ec.I_szy)+spatial.invF31*(spatial.invF11*ec.I_sxz+spatial.invF21*ec.I_syz+spatial.invF31*ec.I_szz))*spatial.detF | 1 | Spatial identity matrix, material frame, XX component | Domains 1–5 |
| ec.I_sYX | (spatial.invF11*(spatial.invF12*ec.I_sxx+spatial.invF22*ec.I_syx+spatial.invF32*ec.I_szx)+spatial.invF21*(spatial.invF12*ec.I_sxy+spatial.invF22*ec.I_syy+spatial.invF32*ec.I_szy)+spatial.invF31*(spatial.invF12*ec.I_sxz+spatial.invF22*ec.I_syz+spatial.invF32*ec.I_szz))*spatial.detF | 1 | Spatial identity matrix, material frame, YX component | Domains 1–5 |
| ec.I_sZX | (spatial.invF11*(spatial.invF13*ec.I_sxx+spatial.invF23*ec.I_syx+spatial.invF33*ec.I_szx)+spatial.invF21*(spatial.invF13*ec.I_sxy+spatial.invF23*ec.I_syy+spatial.invF33*ec.I_szy)+spatial.invF31*(spatial.invF13*ec.I_sxz+spatial.invF23*ec.I_syz+spatial.invF33*ec.I_szz))*spatial.detF | 1 | Spatial identity matrix, material frame, ZX component | Domains 1–5 |
| ec.I_sXY | (spatial.invF12*(spatial.invF11*ec.I_sxx+spatial.invF21*ec.I_syx+spatial.invF31*ec.I_szx)+spatial.invF22*(spatial.invF11*ec.I_sxy+spatial.invF21*ec.I_syy+spatial.invF31*ec.I_szy)+spatial.invF32*(spatial.invF11*ec.I_sxz+spatial.invF21*ec.I_syz+spatial.invF31*ec.I_szz))*spatial.detF | 1 | Spatial identity matrix, material frame, XY component | Domains 1–5 |
| ec.I_sYY | (spatial.invF12*(spatial.invF12*ec.I_sxx+spatial.invF22*ec.I_syx+spatial.invF32*ec.I_szx)+spatial.invF22*(spatial.invF12*ec.I_sxy+spatial.invF22*ec.I_syy+spatial.invF32*ec.I_szy)+spatial.invF32*(spatial.invF12*ec.I_sxz+spatial.invF22*ec.I_syz+spatial.invF32*ec.I_szz))*spatial.detF | 1 | Spatial identity matrix, material frame, YY component | Domains 1–5 |
| ec.I_sZY | (spatial.invF12*(spatial.invF13*ec.I_sxx+spatial.invF23*ec.I_syx+spatial.invF33*ec.I_szx)+spatial.invF22*(spatial.invF13*ec.I_sxy+spatial.invF23*ec.I_syy+spatial.invF33*ec.I_szy)+spatial.invF32*(spatial.invF13*ec.I_sxz+spatial.invF23*ec.I_syz+spatial.invF33*ec.I_szz))*spatial.detF | 1 | Spatial identity matrix, material frame, ZY component | Domains 1–5 |

| | | | | |
|---|---|---|---|---|
| ec.I_sXZ | (spatial.invF13*(spatial.invF11*ec.I_sxx+spatial.invF21*ec.I_syx+spatial.invF31*ec.I_szx)+spatial.invF23*(spatial.invF11*ec.I_sxy+spatial.invF21*ec.I_syy+spatial.invF31*ec.I_szy)+spatial.invF33*(spatial.invF11*ec.I_sxz+spatial.invF21*ec.I_syz+spatial.invF31*ec.I_szz))*spatial.detF | 1 | Spatial identity matrix, material frame, XZ component | Domains 1–5 |
| ec.I_sYZ | (spatial.invF13*(spatial.invF12*ec.I_sxx+spatial.invF22*ec.I_syx+spatial.invF32*ec.I_szx)+spatial.invF23*(spatial.invF12*ec.I_sxy+spatial.invF22*ec.I_syy+spatial.invF32*ec.I_szy)+spatial.invF33*(spatial.invF12*ec.I_sxz+spatial.invF22*ec.I_syz+spatial.invF32*ec.I_szz))*spatial.detF | 1 | Spatial identity matrix, material frame, YZ component | Domains 1–5 |
| ec.I_sZZ | (spatial.invF13*(spatial.invF13*ec.I_sxx+spatial.invF23*ec.I_syx+spatial.invF33*ec.I_szx)+spatial.invF23*(spatial.invF13*ec.I_sxy+spatial.invF23*ec.I_syy+spatial.invF33*ec.I_szy)+spatial.invF33*(spatial.invF13*ec.I_sxz+spatial.invF23*ec.I_syz+spatial.invF33*ec.I_szz))*spatial.detF | 1 | Spatial identity matrix, material frame, ZZ component | Domains 1–5 |
| ec.I_sxx | 1 | 1 | Spatial identity matrix, xx component | Domains 1–5 |
| ec.I_syx | 0 | 1 | Spatial identity matrix, yx component | Domains 1–5 |
| ec.I_szx | 0 | 1 | Spatial identity matrix, zx component | Domains 1–5 |
| ec.I_sxy | 0 | 1 | Spatial identity matrix, xy component | Domains 1–5 |
| ec.I_syy | 1 | 1 | Spatial identity matrix, yy component | Domains 1–5 |
| ec.I_szy | 0 | 1 | Spatial identity matrix, zy component | Domains 1–5 |
| ec.I_sxz | 0 | 1 | Spatial identity matrix, xz component | Domains 1–5 |
| ec.I_syz | 0 | 1 | Spatial identity matrix, yz component | Domains 1–5 |
| ec.I_szz | 1 | 1 | Spatial identity matrix, zz component | Domains 1–5 |
| ec.nx | nx | | Normal vector, x component | Boundaries 2–28 |
| ec.ny | ny | | Normal vector, y component | Boundaries 2–28 |
| ec.nz | nz | | Normal vector, z component | Boundaries 2–28 |
| ec.nx | dnx | | Normal vector, x component | Boundary 1 |
| ec.ny | dny | | Normal vector, y component | Boundary 1 |
| ec.nz | dnz | | Normal vector, z component | Boundary 1 |
| ec.nmeshx | nxmesh | | Mesh normal vector, x component | Boundaries 2–28 |
| ec.nmeshy | nymesh | | Mesh normal vector, y component | Boundaries 2–28 |
| ec.nmeshz | nzmesh | | Mesh normal vector, z component | Boundaries 2–28 |
| ec.nmeshx | dnxmesh | | Mesh normal vector, x component | Boundary 1 |
| ec.nmeshy | dnymesh | | Mesh normal vector, y component | Boundary 1 |
| ec.nmeshz | dnzmesh | | Mesh normal vector, z component | Boundary 1 |
| ec.unmeshx | unxmesh | | Mesh normal vector, upside, x component | Boundaries 1–28 |
| ec.unmeshy | unymesh | | Mesh normal vector, upside, y component | Boundaries 1–28 |
| ec.unmeshz | unzmesh | | Mesh normal vector, upside, z component | Boundaries 1–28 |
| ec.dnmeshx | dnxmesh | | Mesh normal vector, downside, x component | Boundaries 1–28 |
| ec.dnmeshy | dnymesh | | Mesh normal vector, downside, y component | Boundaries 1–28 |
| ec.dnmeshz | dnzmesh | | Mesh normal vector, downside, z component | Boundaries 1–28 |
| ec.unTx | ec.unTex | Pa | Maxwell upward surface stress tensor, x component | Boundaries 1–28 |
| ec.unTy | ec.unTey | Pa | Maxwell upward surface stress tensor, y component | Boundaries 1–28 |
| ec.unTz | ec.unTez | Pa | Maxwell upward surface stress tensor, z component | Boundaries 1–28 |
| ec.dnTx | ec.dnTex | Pa | | |

| | | | | |
|---|---|---|---|---|
| | | | Maxwell downward surface stress tensor, x component | Boundaries 1–28 |
| ec.dnTy | ec.dnTey | Pa | Maxwell downward surface stress tensor, y component | Boundaries 1–28 |
| ec.dnTz | ec.dnTez | Pa | Maxwell downward surface stress tensor, z component | Boundaries 1–28 |
| ec.unx | unx | | Normal vector up direction, x component | Boundaries 1–28 |
| ec.uny | uny | | Normal vector up direction, y component | Boundaries 1–28 |
| ec.unz | unz | | Normal vector up direction, z component | Boundaries 1–28 |
| ec.dnx | dnx | | Normal vector down direction, x component | Boundaries 1–28 |
| ec.dny | dny | | Normal vector down direction, y component | Boundaries 1–28 |
| ec.dnz | dnz | | Normal vector down direction, z component | Boundaries 1–28 |
| ec.unTex | -0.5*ec.dnx*(real(up(ec.Dx))*real(up(ec.Ex))+real(up(ec.Dy))*real(up(ec.Ey))+real(up(ec.Dz))*real(up(ec.Ez)))+real(up(ec.Dx))*(real(up(ec.Ex))*ec.dnx+real(up(ec.Ey))*ec.dny+real(up(ec.Ez))*ec.dnz) | Pa | Maxwell upward electric surface stress tensor, x component | Boundaries 2–28 |
| ec.unTey | -0.5*ec.dny*(real(up(ec.Dx))*real(up(ec.Ex))+real(up(ec.Dy))*real(up(ec.Ey))+real(up(ec.Dz))*real(up(ec.Ez)))+real(up(ec.Dy))*(real(up(ec.Ex))*ec.dnx+real(up(ec.Ey))*ec.dny+real(up(ec.Ez))*ec.dnz) | Pa | Maxwell upward electric surface stress tensor, y component | Boundaries 2–28 |
| ec.unTez | -0.5*ec.dnz*(real(up(ec.Dx))*real(up(ec.Ex))+real(up(ec.Dy))*real(up(ec.Ey))+real(up(ec.Dz))*real(up(ec.Ez)))+real(up(ec.Dz))*(real(up(ec.Ex))*ec.dnx+real(up(ec.Ey))*ec.dny+real(up(ec.Ez))*ec.dnz) | Pa | Maxwell upward electric surface stress tensor, z component | Boundaries 2–28 |
| ec.unTex | 0 | Pa | Maxwell upward electric surface stress tensor, x component | Boundary 1 |
| ec.unTey | 0 | Pa | Maxwell upward electric surface stress tensor, y component | Boundary 1 |
| ec.unTez | 0 | Pa | Maxwell upward electric surface stress tensor, z component | Boundary 1 |
| ec.dnTex | -0.5*ec.unx*(real(down(ec.Dx))*real(down(ec.Ex))+real(down(ec.Dy))*real(down(ec.Ey))+real(down(ec.Dz))*real(down(ec.Ez)))+real(down(ec.Dx))*(real(down(ec.Ex))*ec.unx+real(down(ec.Ey))*ec.uny+real(down(ec.Ez))*ec.unz) | Pa | Maxwell downward electric surface stress tensor, x component | Boundaries 1–28 |
| ec.dnTey | -0.5*ec.uny*(real(down(ec.Dx))*real(down(ec.Ex))+real(down(ec.Dy))*real(down(ec.Ey))+real(down(ec.Dz))*real(down(ec.Ez)))+real(down(ec.Dy))*(real(down(ec.Ex))*ec.unx+real(down(ec.Ey))*ec.uny+real(down(ec.Ez))*ec.unz) | Pa | Maxwell downward electric surface stress tensor, y component | Boundaries 1–28 |
| ec.dnTez | -0.5*ec.unz*(real(down(ec.Dx))*real(down(ec.Ex))+real(down(ec.Dy))*real(down(ec.Ey))+real(down(ec.Dz))*real(down(ec.Ez)))+real(down(ec.Dz))*(real(down(ec.Ex))*ec.unx+real(down(ec.Ey))*ec.uny+real(down(ec.Ez))*ec.unz) | Pa | Maxwell downward electric surface stress tensor, z component | Boundaries 1–28 |
| ec.intWe | ec.int_We(ec.d*ec.dWe) | J | Total electric energy | Global |
| ec.Qh | 0 | W/m³ | Volumetric loss density, electromagnetic | Domains 1–5 |
| ec.Qsh | 0 | W/m² | Surface loss density, electromagnetic | Boundaries 1–28 |
| ec.Qlh | 0 | W/m | Line loss density, electromagnetic | Edges 1–52 |
| ec.zref | 50[ohm] | Ω | Reference impedance | Global |

**2.4.3. Current Conservation 1**




*Current Conservation 1*

Selection

| | |
|---|---|
| Geometric entity level | Domain |
| Selection | Geometry geom1: Dimension 3: All domains |

Equations

$$\nabla \cdot \mathbf{J} = Q_{j,v}$$

$$\mathbf{J} = \sigma \mathbf{E} + \mathbf{J}_e$$

$$\mathbf{E} = -\nabla V$$

**Constitutive Relation Jc-E**

Settings

| Description | Value |
|---|---|
| Conduction model | Electrical conductivity |
| Electrical conductivity | From material |

**Constitutive Relation D-E**

Settings

| Description | Value |
|---|---|
| Dielectric model | Relative permittivity |
| Relative permittivity | From material |

**Coordinate System Selection**

Settings

| Description | Value |
|---|---|
| Coordinate system | Global coordinate system |

Properties from material

| Property | Material | Property group |
|---|---|---|
| Electrical conductivity | Liver (human) | Basic |
| Relative permittivity | Liver (human) | Basic |
| Electrical conductivity | Blood | Basic |
| Relative permittivity | Blood | Basic |
| Electrical conductivity | Electrodes | Basic |
| Relative permittivity | Electrodes | Basic |
| Electrical conductivity | Trocar Tip | Basic |
| Relative permittivity | Trocar Tip | Basic |
| Electrical conductivity | Trocar Base | Basic |
| Relative permittivity | Trocar Base | Basic |

**Variables**

| Name | Expression | Unit | Description | Selection | Detail |
|---|---|---|---|---|---|

| | | | | | |
|---|---|---|---|---|---|
| ec.Qh | ec.Qrh | W/m³ | Volumetric loss density, electromagnetic | Domains 1–5 | |
| ec.Jix | ec.sigmaxx*ec.Ex+ec.sigmaxy*ec.Ey+ec.sigmaxz*ec.Ez | A/m² | Conduction current density, x component | Domains 1–5 | |
| ec.Jiy | ec.sigmayx*ec.Ex+ec.sigmayy*ec.Ey+ec.sigmayz*ec.Ez | A/m² | Conduction current density, y component | Domains 1–5 | |
| ec.Jiz | ec.sigmazx*ec.Ex+ec.sigmazy*ec.Ey+ec.sigmazz*ec.Ez | A/m² | Conduction current density, z component | Domains 1–5 | |
| ec.Jdx | 0 | A/m² | Displacement current density, x component | Domains 1–5 | |
| ec.Jdy | 0 | A/m² | Displacement current density, y component | Domains 1–5 | |
| ec.Jdz | 0 | A/m² | Displacement current density, z component | Domains 1–5 | |
| ec.Jex | 0 | A/m² | External current density, x component | Domains 1–5 | + operati |
| ec.Jey | 0 | A/m² | External current density, y component | Domains 1–5 | + operati |
| ec.Jez | 0 | A/m² | External current density, z component | Domains 1–5 | + operati |
| ec.Jx | ec.Jix+ec.Jdx+ec.Jex | A/m² | Current density, x component | Domains 1–5 | |
| ec.Jy | ec.Jiy+ec.Jdy+ec.Jey | A/m² | Current density, y component | Domains 1–5 | |
| ec.Jz | ec.Jiz+ec.Jdz+ec.Jez | A/m² | Current density, z component | Domains 1–5 | |
| ec.normJ | sqrt(realdot(ec.Jx,ec.Jx)+realdot(ec.Jy,ec.Jy)+realdot(ec.Jz,ec.Jz)) | A/m² | Current density norm | Domains 1–5 | |
| ec.rhoq | ppr(d(ec.Dx,x)+d(ec.Dy,y)+d(ec.Dz,z)) | C/m³ | Space charge density | Domains 1–5 | |
| ec.sigmaxx | material.sigma11 | S/m | Electrical conductivity, xx component | Domains 1–5 | Meta |
| ec.sigmayx | material.sigma21 | S/m | Electrical conductivity, yx component | Domains 1–5 | Meta |
| ec.sigmazx | material.sigma31 | S/m | Electrical conductivity, zx component | Domains 1–5 | Meta |
| ec.sigmaxy | material.sigma12 | S/m | Electrical conductivity, xy component | Domains 1–5 | Meta |
| ec.sigmayy | material.sigma22 | S/m | Electrical conductivity, yy component | Domains 1–5 | Meta |
| ec.sigmazy | material.sigma32 | S/m | Electrical conductivity, zy component | Domains 1–5 | Meta |
| ec.sigmaxz | material.sigma13 | S/m | Electrical conductivity, xz component | Domains 1–5 | Meta |
| ec.sigmayz | material.sigma23 | S/m | Electrical conductivity, yz component | Domains 1–5 | Meta |
| ec.sigmazz | material.sigma33 | S/m | Electrical conductivity, zz component | Domains 1–5 | Meta |
| ec.sigma_iso | material.sigma_iso | S/m | Electrical conductivity, isotropic value | Domains 1–5 | Meta |
| ec.epsilonrxx | material.epsilonr11 | 1 | | Domains 1–5 | Meta |

| | | | | | |
|---|---|---|---|---|---|
| | | | Relative permittivity, xx component | | |
| ec.epsilonryx | material.epsilonr21 | 1 | Relative permittivity, yx component | Domains 1–5 | Meta |
| ec.epsilonrzx | material.epsilonr31 | 1 | Relative permittivity, zx component | Domains 1–5 | Meta |
| ec.epsilonrxy | material.epsilonr12 | 1 | Relative permittivity, xy component | Domains 1–5 | Meta |
| ec.epsilonryy | material.epsilonr22 | 1 | Relative permittivity, yy component | Domains 1–5 | Meta |
| ec.epsilonrzy | material.epsilonr32 | 1 | Relative permittivity, zy component | Domains 1–5 | Meta |
| ec.epsilonrxz | material.epsilonr13 | 1 | Relative permittivity, xz component | Domains 1–5 | Meta |
| ec.epsilonryz | material.epsilonr23 | 1 | Relative permittivity, yz component | Domains 1–5 | Meta |
| ec.epsilonrzz | material.epsilonr33 | 1 | Relative permittivity, zz component | Domains 1–5 | Meta |
| ec.epsilonr_iso | material.epsilonr_iso | 1 | Relative permittivity, isotropic value | Domains 1–5 | Meta |
| ec.Dx | epsilon0_const*ec.I_sxx*ec.Ex+epsilon0_const*ec.I_sxy*ec.Ey+epsilon0_const*ec.I_sxz*ec.Ez+ec.Px+ec.Pex | C/m² | Electric displacement field, x component | Domains 1–5 | |
| ec.Dy | epsilon0_const*ec.I_syx*ec.Ex+epsilon0_const*ec.I_syy*ec.Ey+epsilon0_const*ec.I_syz*ec.Ez+ec.Py+ec.Pey | C/m² | Electric displacement field, y component | Domains 1–5 | |
| ec.Dz | epsilon0_const*ec.I_szx*ec.Ex+epsilon0_const*ec.I_szy*ec.Ey+epsilon0_const*ec.I_szz*ec.Ez+ec.Pz+ec.Pez | C/m² | Electric displacement field, z component | Domains 1–5 | |
| ec.Px | epsilon0_const*(ec.chixx*ec.Ex+ec.chixy*ec.Ey+ec.chixz*ec.Ez) | C/m² | Polarization, x component | Domains 1–5 | |
| ec.Py | epsilon0_const*(ec.chiyx*ec.Ex+ec.chiyy*ec.Ey+ec.chiyz*ec.Ez) | C/m² | Polarization, y component | Domains 1–5 | |
| ec.Pz | epsilon0_const*(ec.chizx*ec.Ex+ec.chizy*ec.Ey+ec.chizz*ec.Ez) | C/m² | Polarization, z component | Domains 1–5 | |
| ec.normD | sqrt(realdot(ec.Dx,ec.Dx)+realdot(ec.Dy,ec.Dy)+realdot(ec.Dz,ec.Dz)) | C/m² | Electric displacement field norm | Domains 1–5 | |
| ec.normP | sqrt(realdot(ec.Px,ec.Px)+realdot(ec.Py,ec.Py)+realdot(ec.Pz,ec.Pz)) | C/m² | Polarization norm | Domains 1–5 | |
| ec.Pex | 0 | C/m² | Polarization contribution, x component | Domains 1–5 | + operati |
| ec.Pey | 0 | C/m² | Polarization contribution, y component | Domains 1–5 | + operati |
| ec.Pez | 0 | C/m² | Polarization contribution, z component | Domains 1–5 | + operati |
| ec.chixx | -1+ec.epsilonrxx | 1 | Electric susceptibility, xx component | Domains 1–5 | |
| ec.chiyx | ec.epsilonryx | 1 | Electric susceptibility, yx component | Domains 1–5 | |
| ec.chizx | ec.epsilonrzx | 1 | Electric susceptibility, zx component | Domains 1–5 | |
| ec.chixy | ec.epsilonrxy | 1 | Electric susceptibility, xy component | Domains 1–5 | |
| ec.chiyy | -1+ec.epsilonryy | 1 | | | |

| | | | | | |
|---|---|---|---|---|---|
| | | | Electric susceptibility, yy component | Domains 1–5 | |
| ec.chizy | ec.epsilonrzy | 1 | Electric susceptibility, zy component | Domains 1–5 | |
| ec.chixz | ec.epsilonrxz | 1 | Electric susceptibility, xz component | Domains 1–5 | |
| ec.chiyz | ec.epsilonryz | 1 | Electric susceptibility, yz component | Domains 1–5 | |
| ec.chizz | -1+ec.epsilonrzz | 1 | Electric susceptibility, zz component | Domains 1–5 | |
| ec.Ex | -Vx | V/m | Electric field, x component | Domains 1–5 | |
| ec.Ey | -Vy | V/m | Electric field, y component | Domains 1–5 | |
| ec.Ez | -Vz | V/m | Electric field, z component | Domains 1–5 | |
| ec.tEx | -VTx | V/m | Tangential electric field, x component | Boundaries 1–28 | |
| ec.tEy | -VTy | V/m | Tangential electric field, y component | Boundaries 1–28 | |
| ec.tEz | -VTz | V/m | Tangential electric field, z component | Boundaries 1–28 | |
| ec.normE | sqrt(realdot(ec.Ex,ec.Ex)+realdot(ec.Ey,ec.Ey)+realdot(ec.Ez,ec.Ez)) | V/m | Electric field norm | Domains 1–5 | |
| ec.Qrh | ec.Jx*ec.Ex+ec.Jy*ec.Ey+ec.Jz*ec.Ez | W/m³ | Volumetric loss density, electric | Domains 1–5 | + operati |
| ec.W | ec.We | J/m³ | Energy density | Domains 1–5 | + operati |
| ec.dWe | ec.We | J/m³ | Integrand for total electric energy | Domains 1–5 | Meta |
| ec.We | 0.5*epsilon0_const*(((ec.I_sxx+ec.chixx)*ec.Ex+(ec.I_sxy+ec.chixy)*ec.Ey+(ec.I_sxz+ec.chixz)*ec.Ez)*ec.Ex+((ec.I_syx+ec.chiyx)*ec.Ex+(ec.I_syy+ec.chiyy)*ec.Ey+(ec.I_syz+ec.chiyz)*ec.Ez)*ec.Ey+((ec.I_szx+ec.chizx)*ec.Ex+(ec.I_szy+ec.chizy)*ec.Ey+(ec.I_szz+ec.chizz)*ec.Ez)*ec.Ez) | J/m³ | Electric energy density | Domains 1–5 | |
| ec.rhoqs | ec.nx*(up(ec.Dx)-down(ec.Dx))+ec.ny*(up(ec.Dy)-down(ec.Dy))+ec.nz*(up(ec.Dz)-down(ec.Dz)) | C/m² | Surface charge density | Boundaries 2–28 | |
| ec.rhoqs | -ec.nx*down(ec.Dx)-ec.ny*down(ec.Dy)-ec.nz*down(ec.Dz) | C/m² | Surface charge density | Boundary 1 | |

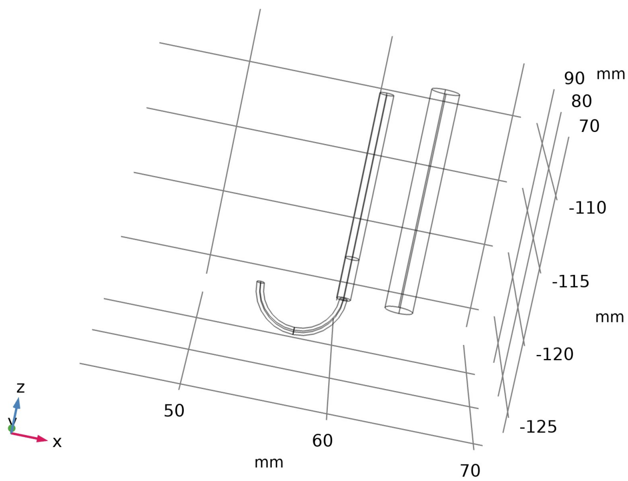
**Shape functions**

| Name | Shape function | Unit | Description | Shape frame | Selection |
|---|---|---|---|---|---|
| V | Lagrange (Linear) | V | Electric potential | Spatial | Domains 1–5 |
| V | Lagrange (Linear) | V | Electric potential | Material | Domains 1–5 |
| V | Lagrange (Linear) | V | Electric potential | Geometry | Domains 1–5 |
| V | Lagrange (Linear) | V | Electric potential | Mesh | Domains 1–5 |

**Weak Expressions**

| Weak expression | Integration order | Integration frame | Selection |
|---|---|---|---|
| (ec.Jx*test(Vx)+ec.Jy*test(Vy)+ec.Jz*test(Vz))*ec.d | 2 | Spatial | Domains 1–5 |

**2.4.4. Electric Insulation 1**




*Electric Insulation 1*

Selection

| | |
|---|---|
| Geometric entity level | Boundary |
| Selection | Geometry geom1: Dimension 2: All boundaries |

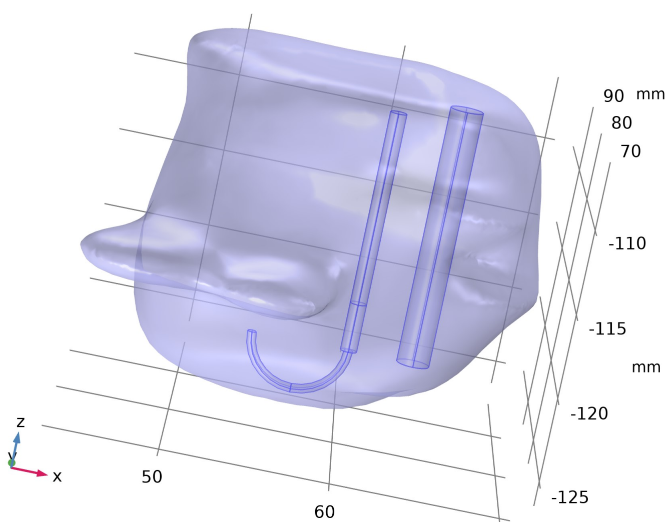
Equations

$$\mathbf{n}\cdot\mathbf{J}=0$$

**Shape functions**

| Name | Shape function | Unit | Description | Shape frame | Selection | Details |
|---|---|---|---|---|---|---|
| V | Lagrange (Linear) | V | Electric potential | Spatial | No boundaries | Slit |
| V | Lagrange (Linear) | V | Electric potential | Material | No boundaries | Slit |
| V | Lagrange (Linear) | V | Electric potential | Geometry | No boundaries | Slit |
| V | Lagrange (Linear) | V | Electric potential | Mesh | No boundaries | Slit |

**2.4.5. Initial Values 1**




*Initial Values 1*

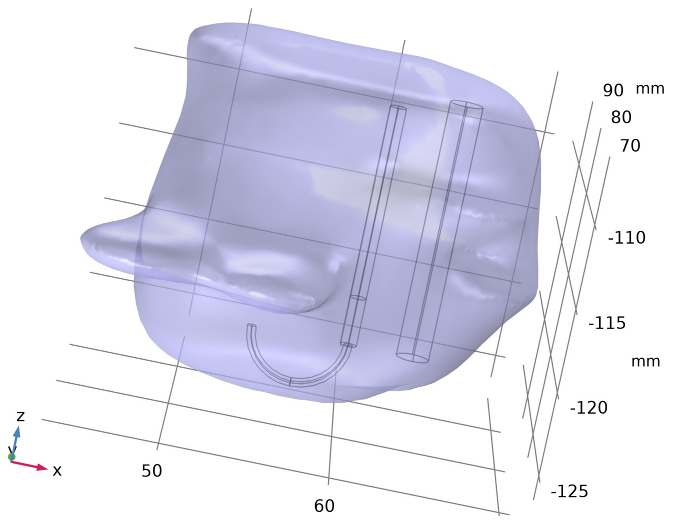
Selection

| | |
|---|---|
| Geometric entity level | Domain |
| Selection | Geometry geom1: Dimension 3: All domains |

Settings

| Description | Value |
|---|---|
| Electric potential | 0 |

#### 2.4.6. Ground 1




*Ground 1*

Selection

| | |
|---|---|
| Geometric entity level | Boundary |
| Name | Exterior Boundaries |
| Selection | Named sel1: Geometry geom1: Input dimension 3 Output [EXTERIOR]: Domains 1–5; Dimension 2: Boundary 1 |

Equations

$$V = 0$$

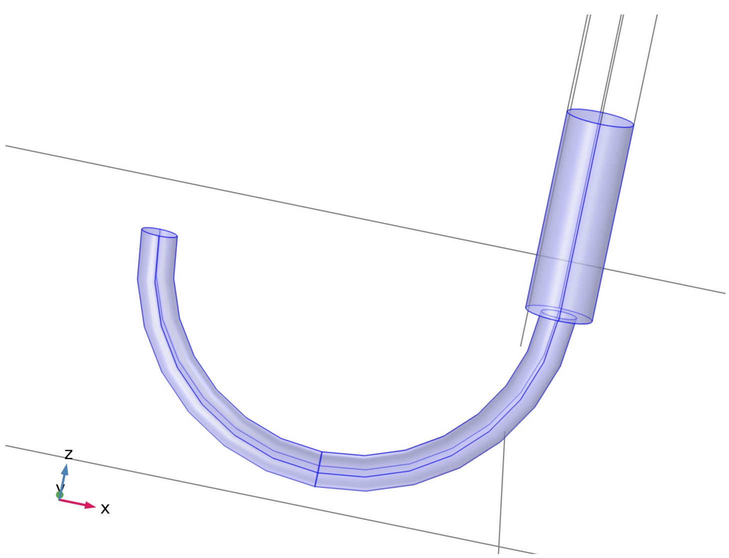
##### Variables

| Name | Expression | Unit | Description | Selection | Details |
|---|---|---|---|---|---|
| ec.nJ | ec.unx*down(ec.Jx)+ec.uny*down(ec.Jy)+ec.unz*down(ec.Jz) | A/m² | Normal current density | Boundary 1 | + operation |
| ec.V0 | 0 | V | Electric potential | Boundary 1 | |

##### Constraints

| Constraint | Constraint force | Shape function | Selection | Details |
|---|---|---|---|---|
| ec.V0-V | test(ec.V0-V) | Lagrange (Linear) | Boundary 1 | Elemental |

#### 2.4.7. Electric Potential 1




*Electric Potential 1*

Selection

| | |
|---|---|
| Geometric entity level | Boundary |
| Name | Trocar Tip and Electrodes, Exterior boundaries |
| Selection | Named adj2: Geometry geom1: Input dimension 3 Output [EXTERIOR]: Domains 2–3; Dimension 2: Boundaries 2–13, 16, 19, 21 |

Equations

$$V = V_0$$

##### Electric Potential

Settings

| Description | Value |
|---|---|
| Electric potential | V0 |

##### Variables

| Name | Expression | Unit | Description | Selection | Details |
|---|---|---|---|---|---|
| ec.nJ | ec.unx*(down(ec.Jx)-up(ec.Jx))+ec.uny*(down(ec.Jy)-up(ec.Jy))+ec.unz*(down(ec.Jz)-up (ec.Jz)) | $A/m^2$ | Normal current density | Boundaries 2–13, 16, 19, 21 | + operation |
| ec.V0 | V0 | V | Electric potential | Boundaries 2–13, 16, 19, 21 | |

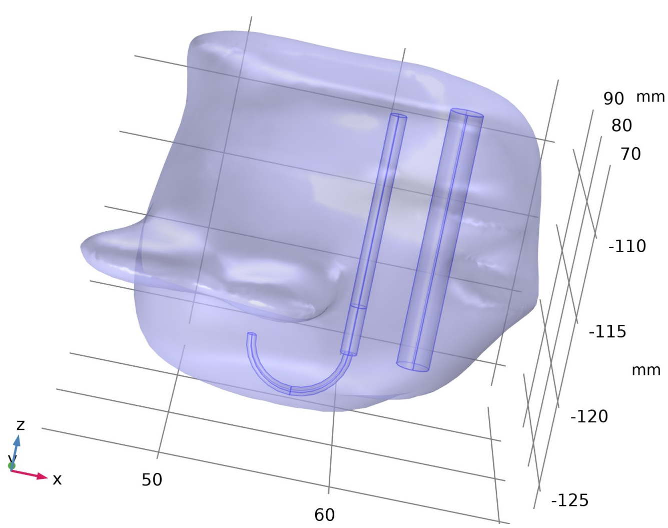
##### Constraints

| Constraint | Constraint force | Shape function | Selection | Details |
|---|---|---|---|---|
| ec.V0-V | test(ec.V0-V) | Lagrange (Linear) | Boundaries 2–13, 16, 19, 21 | Elemental |

### 2.5. Bioheat Transfer

Used products

| |
|---|
| COMSOL Multiphysics |
| Heat Transfer Module |




*Bioheat Transfer*

Selection

| | |
|---|---|
| Geometric entity level | Domain |
| Name | Tissue and Trocar |
| Selection | Named uni2: Geometry geom1: Dimension 3: Domains 1–4 |

Equations

$$\rho C_p \frac{\partial T}{\partial t} + \rho C_p \mathbf{u} \cdot \nabla T + \nabla \cdot \mathbf{q} = Q + Q_{\text{bio}}$$

$$\mathbf{q} = -k \nabla T$$

#### 2.5.1. Interface Settings

##### Discretization

Settings

| Description | Value |
| --- | --- |
| Temperature | Quadratic Lagrange |
| Damaged tissue indicator | Constant |

###### Physical Model

Settings

| Description | Value |
| --- | --- |
| Isothermal domain | Off |
| Reference temperature | User defined |
| Reference temperature | 293.15[K] |

#### 2.5.2. Variables

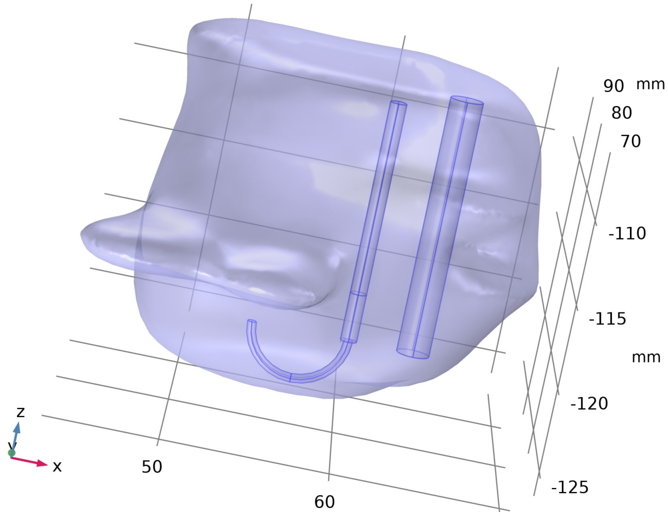
| Name | Expression | Unit | Description | Selection | Details |
| --- | --- | --- | --- | --- | --- |
| ht.ndflux_u | -ht.ndflux_d | W/m² | Internal normal conductive heat flux, upside | Boundary 1 | + operation |
| ht.ndflux_u | 0 | W/m² | Internal normal conductive heat flux, upside | Boundaries 2–28 | + operation |
| ht.ndflux_d | -ht.ndflux_u | W/m² | Internal normal conductive heat flux, downside | Boundaries 23–28 | + operation |
| ht.ndflux_d | 0 | W/m² | Internal normal conductive heat flux, downside | Boundaries 1–22 | + operation |
| ht.Q | 0 | W/m³ | Heat source | Domains 1–4 | + operation |
| ht.Qtot | 0 | W/m³ | Total heat source | Domains 1–4 | + operation |
| ht.Qbtot | 0 | W/m² | Total boundary heat source | Boundaries 1–28 | + operation |
| ht.ntflux_contrib | 0 | W/m² | Boundary sources and fluxes contribution | Domains 1–4 | + operation |
| ht.Tref | model.input.Tref | K | Reference temperature | Global | Meta |
| ht.d | 1 | 1 | Thickness | Domains 1–4 | |
| ht.HRef | 0 | J/kg | Reference enthalpy | Domains 1–4 | |
| ht.alphap | 0 | 1/K | Isobaric compressibility coefficient | Domains 1–4 | |
| ht.DeltaH | 0 | J/kg | Sensible enthalpy | Domains 1–4 | + operation |
| ht.H | 0 | J/kg | Enthalpy | Domains 1–4 | + operation |
| ht.H0 | ht.H+ht.Ek | J/kg | Total enthalpy | Domains 1–4 | |
| ht.Ei | 0 | J/kg | Internal energy | Domains 1–4 | + operation |
| ht.Ei0 | ht.Ei+ht.Ek | J/kg | Total internal energy | Domains 1–4 | |
| ht.Ek | 0 | J/kg | Kinetic energy | Domains 1–4 | + operation |
| ht.dfluxx | 0 | W/m² | Conductive heat flux, x component | Domains 1–4 | + operation |
| ht.dfluxy | 0 | W/m² | Conductive heat flux, y component | Domains 1–4 | + operation |
| ht.dfluxz | 0 | W/m² | Conductive heat flux, z component | Domains 1–4 | + operation |
| ht.dfluxx | mean(ht.dfluxx) | W/m² | Conductive heat flux, x component | Boundaries 1–28 | + operation |
| ht.dfluxy | mean(ht.dfluxy) | W/m² | Conductive heat flux, y component | Boundaries 1–28 | + operation |
| ht.dfluxz | mean(ht.dfluxz) | W/m² | Conductive heat flux, z component | Boundaries 1–28 | + operation |
| ht.dfluxtestx | 0 | W/m² | Conductive heat flux, x component | Domains 1–4 | + operation |
| ht.dfluxtesty | 0 | W/m² | Conductive heat flux, y component | Domains 1–4 | + operation |
| ht.dfluxtestz | 0 | W/m² | Conductive heat flux, z component | Domains 1–4 | + operation |
| ht.dfluxtestx | mean(ht.dfluxtestx) | W/m² | Conductive heat flux, x component | Boundaries 1–28 | + operation |
| ht.dfluxtesty | mean(ht.dfluxtesty) | W/m² | Conductive heat flux, y component | Boundaries 1–28 | + operation |
| ht.dfluxtestz | mean(ht.dfluxtestz) | W/m² | Conductive heat flux, z component | Boundaries 1–28 | + operation |
| ht.dfluxMag | sqrt(ht.dfluxx^2+ht.dfluxy^2+ht.dfluxz^2) | W/m² | Conductive heat flux magnitude | Domains 1–4 | |
| ht.cfluxx | 0 | W/m² | Convective heat flux, x component | Domains 1–4 | + operation |
| ht.cfluxy | 0 | W/m² | | Domains 1–4 | + operation |

| | | | | | |
|---|---|---|---|---|---|
| | | | Convective heat flux, y component | | |
| ht.cfluxz | 0 | W/m² | Convective heat flux, z component | Domains 1–4 | + operation |
| ht.cfluxMag | sqrt(ht.cfluxx^2+ht.cfluxy^2+ht.cfluxz^2) | W/m² | Convective heat flux magnitude | Domains 1–4 | |
| ht.tfluxx | ht.dfluxx+ht.cfluxx | W/m² | Total heat flux, x component | Domains 1–4 | |
| ht.tfluxy | ht.dfluxy+ht.cfluxy | W/m² | Total heat flux, y component | Domains 1–4 | |
| ht.tfluxz | ht.dfluxz+ht.cfluxz | W/m² | Total heat flux, z component | Domains 1–4 | |
| ht.tfluxMag | sqrt(ht.tfluxx^2+ht.tfluxy^2+ht.tfluxz^2) | W/m² | Total heat flux magnitude | Domains 1–4 | |
| ht.tefluxx | 0 | W/m² | Total energy flux, x component | Domains 1–4 | + operation |
| ht.tefluxy | 0 | W/m² | Total energy flux, y component | Domains 1–4 | + operation |
| ht.tefluxz | 0 | W/m² | Total energy flux, z component | Domains 1–4 | + operation |
| ht.tefluxMag | sqrt(ht.tefluxx^2+ht.tefluxy^2+ht.tefluxz^2) | W/m² | Total energy flux magnitude | Domains 1–4 | |
| ht.thfluxx | 0 | W/m² | Total enthalpy flux, x component | Domains 1–4 | + operation |
| ht.thfluxy | 0 | W/m² | Total enthalpy flux, y component | Domains 1–4 | + operation |
| ht.thfluxz | 0 | W/m² | Total enthalpy flux, z component | Domains 1–4 | + operation |
| ht.thfluxMag | sqrt(ht.thfluxx^2+ht.thfluxy^2+ht.thfluxz^2) | W/m² | Total enthalpy flux magnitude | Domains 1–4 | |
| ht.dflux_ux | up(ht.dfluxx) | W/m² | Conductive heat flux, x component | Boundaries 1–28 | |
| ht.dflux_uy | up(ht.dfluxy) | W/m² | Conductive heat flux, y component | Boundaries 1–28 | |
| ht.dflux_uz | up(ht.dfluxz) | W/m² | Conductive heat flux, z component | Boundaries 1–28 | |
| ht.dflux_dx | down(ht.dfluxx) | W/m² | Conductive heat flux, x component | Boundaries 1–28 | |
| ht.dflux_dy | down(ht.dfluxy) | W/m² | Conductive heat flux, y component | Boundaries 1–28 | |
| ht.dflux_dz | down(ht.dfluxz) | W/m² | Conductive heat flux, z component | Boundaries 1–28 | |
| ht.dfluxtest_ux | up(ht.dfluxtestx) | W/m² | Conductive heat flux, x component | Boundaries 1–28 | |
| ht.dfluxtest_uy | up(ht.dfluxtesty) | W/m² | Conductive heat flux, y component | Boundaries 1–28 | |
| ht.dfluxtest_uz | up(ht.dfluxtestz) | W/m² | Conductive heat flux, z component | Boundaries 1–28 | |
| ht.dfluxtest_dx | down(ht.dfluxtestx) | W/m² | Conductive heat flux, x component | Boundaries 1–28 | |
| ht.dfluxtest_dy | down(ht.dfluxtesty) | W/m² | Conductive heat flux, y component | Boundaries 1–28 | |
| ht.dfluxtest_dz | down(ht.dfluxtestz) | W/m² | Conductive heat flux, z component | Boundaries 1–28 | |
| ht.rflux | 0 | W/m² | Radiative heat flux | Boundaries 1–28 | + operation |
| ht.ncflux | mean(ht.cfluxx)*ht.nxmesh+mean(ht.cfluxy)*ht.nymesh+mean(ht.cfluxz)*ht.nzmesh | W/m² | Normal convective heat flux | Boundaries 1–28 | |
| ht.ncflux_u | up(ht.cfluxx)*ht.unxmesh+up(ht.cfluxy)*ht.unymesh+up(ht.cfluxz)*ht.unzmesh | W/m² | Internal normal convective heat flux, upside | Boundaries 1–28 | |
| ht.ncflux_d | down(ht.cfluxx)*ht.dnxmesh+down(ht.cfluxy)*ht.dnymesh+down(ht.cfluxz)*ht.dnzmesh | W/m² | Internal normal convective heat flux, downside | Boundaries 1–28 | |
| ht.ndflux | -0.5*(ht.ndflux_d-ht.ndflux_u) | W/m² | Normal conductive heat flux | Boundaries 23–28 | + operation |
| ht.ndflux | 0.5*(ht.ndflux_d-ht.ndflux_u) | W/m² | Normal conductive heat flux | Boundaries 1–22 | + operation |
| ht.ntflux | ht.ndflux+ht.ncflux | W/m² | Normal total heat flux | Boundaries 1–28 | |
| ht.ntflux_u | ht.ndflux_u+ht.ncflux_u | W/m² | Internal normal total flux, upside | Boundaries 1–28 | |
| ht.ntflux_d | ht.ndflux_d+ht.ncflux_d | W/m² | Internal normal total flux, downside | Boundaries 1–28 | |
| ht.nteflux | mean(ht.tefluxx)*ht.nxmesh+mean(ht.tefluxy)*ht.nymesh+mean(ht.tefluxz)*ht.nzmesh-mean(ht.dfluxx)*ht.nxmesh-mean(ht.dfluxy)*ht.nymesh-mean(ht.dfluxz)*ht.nzmesh+ht.ndflux | W/m² | Normal total energy flux | Boundaries 1–28 | |
| ht.nteflux_u | | W/m² | Internal normal total energy flux, upside | Boundaries 1–28 | |

| | | | | | |
|---|---|---|---|---|---|
| | up(ht.tefluxx)*ht.unxmesh+up(ht.tefluxy)*ht.unymesh+up(ht.tefluxz)*ht.unzmesh-up(ht.dfluxx)*ht.unxmesh-up(ht.dfluxy)*ht.unymesh-up(ht.dfluxz)*ht.unzmesh+ht.ndflux_u | | | | |
| ht.nteflux_d | down(ht.tefluxx)*ht.dnxmesh+down(ht.tefluxy)*ht.dnymesh+down(ht.tefluxz)*ht.dnzmesh-down(ht.dfluxx)*ht.dnxmesh-down(ht.dfluxy)*ht.dnymesh-down(ht.dfluxz)*ht.dnzmesh+ht.ndflux_d | W/m² | Internal normal total energy flux, downside | Boundaries 1–28 | |
| ht.nthflux | mean(ht.thfluxx)*ht.nxmesh+mean(ht.thfluxy)*ht.nymesh+mean(ht.thfluxz)*ht.nzmesh | W/m² | Normal total enthalpy flux | Boundaries 1–28 | |
| ht.nthflux_u | up(ht.thfluxx)*ht.unxmesh+up(ht.thfluxy)*ht.unymesh+up(ht.thfluxz)*ht.unzmesh | W/m² | Internal normal total enthalpy flux, upside | Boundaries 1–28 | |
| ht.nthflux_d | down(ht.thfluxx)*ht.dnxmesh+down(ht.thfluxy)*ht.dnymesh+down(ht.thfluxz)*ht.dnzmesh | W/m² | Internal normal total enthalpy flux, downside | Boundaries 1–28 | |
| ht.Qm | 0 | kg/(m³·s) | Mass source | Domains 1–4 | |
| ht.Qoop | 0 | W/m³ | Out-of-plane heat source | Domains 1–4 | + operation |
| ht.Qitot | 0 | W/m² | Total interface source | Domains 1–4 | + operation |
| ht.qs | 0 | W/(m³·K) | Production/absorption coefficient | Domains 1–4 | + operation |
| ht.qs_oop | 0 | W/(m³·K) | Out-of-plane production/absorption coefficient | Domains 1–4 | + operation |
| ht.Qltot | 0 | W/m | Total line heat source | Edges 1–52 | + operation |
| ht.Qlrtot | 0 | W/m | Total line heat source with radius | Edges 1–52 | + operation |
| ht.Qptot | 0 | W | Total point heat source | Points 1–32 | + operation |
| ht.Qprtot | 0 | W | Total point heat source with radius | Points 1–32 | + operation |
| ht.Tvar | T | K | Temperature | Domains 1–4 | |
| ht.Tvar | T | K | Temperature | Boundaries 1–28 | |
| ht.Tvar | T | K | Temperature | Edges 1–52 | |
| ht.Tvar | T | K | Temperature | Points 1–32 | |
| ht.Tu | up(T) | K | Temperature | Boundaries 2–22 | |
| ht.Tu | T | K | Temperature | Boundaries 1, 23–28 | |
| ht.Td | down(T) | K | Temperature | Boundaries 2–22 | |
| ht.Td | T | K | Temperature | Boundaries 1, 23–28 | |
| ht.du | up(ht.d) | 1 | Thickness | Boundaries 2–22 | |
| ht.du | ht.d | 1 | Thickness | Boundaries 1, 23–28 | |
| ht.dd | down(ht.d) | 1 | Thickness | Boundaries 2–22 | |
| ht.dd | ht.d | 1 | Thickness | Boundaries 1, 23–28 | |
| ht.q0 | 0 | W/m² | Inward heat flux | Boundaries 1, 23–28 | + operation |
| ht.nx | nx | 1 | Normal vector, x component | Boundaries 2–22 | |
| ht.ny | ny | 1 | Normal vector, y component | Boundaries 2–22 | |
| ht.nz | nz | 1 | Normal vector, z component | Boundaries 2–22 | |
| ht.nx | unx | 1 | Normal vector, x component | Boundaries 23–28 | |
| ht.ny | uny | 1 | Normal vector, y component | Boundaries 23–28 | |
| ht.nz | unz | 1 | Normal vector, z component | Boundaries 23–28 | |
| ht.nx | dnx | 1 | Normal vector, x component | Boundary 1 | |
| ht.ny | dny | 1 | Normal vector, y component | Boundary 1 | |
| ht.nz | dnz | 1 | Normal vector, z component | Boundary 1 | |
| ht.nxmesh | nxmesh | 1 | Normal vector (mesh), x component | Boundaries 2–22 | |
| ht.nymesh | nymesh | 1 | Normal vector (mesh), y component | Boundaries 2–22 | |
| ht.nzmesh | nzmesh | 1 | Normal vector (mesh), z component | Boundaries 2–22 | |

| | | | | | |
|---|---|---|---|---|---|
| ht.nxmesh | unxmesh | 1 | Normal vector (mesh), x component | Boundaries 23–28 | |
| ht.nymesh | unymesh | 1 | Normal vector (mesh), y component | Boundaries 23–28 | |
| ht.nzmesh | unzmesh | 1 | Normal vector (mesh), z component | Boundaries 23–28 | |
| ht.nxmesh | dnxmesh | 1 | Normal vector (mesh), x component | Boundary 1 | |
| ht.nymesh | dnymesh | 1 | Normal vector (mesh), y component | Boundary 1 | |
| ht.nzmesh | dnzmesh | 1 | Normal vector (mesh), z component | Boundary 1 | |
| ht.dnx | dnx | 1 | Normal vector down direction, x component | Boundaries 1–28 | |
| ht.dny | dny | 1 | Normal vector down direction, y component | Boundaries 1–28 | |
| ht.dnz | dnz | 1 | Normal vector down direction, z component | Boundaries 1–28 | |
| ht.dnxmesh | dnxmesh | 1 | Normal vector down direction (mesh), x component | Boundaries 1–28 | |
| ht.dnymesh | dnymesh | 1 | Normal vector down direction (mesh), y component | Boundaries 1–28 | |
| ht.dnzmesh | dnzmesh | 1 | Normal vector down direction (mesh), z component | Boundaries 1–28 | |
| ht.unx | unx | 1 | Normal vector up direction, x component | Boundaries 1–28 | |
| ht.uny | uny | 1 | Normal vector up direction, y component | Boundaries 1–28 | |
| ht.unz | unz | 1 | Normal vector up direction, z component | Boundaries 1–28 | |
| ht.unxmesh | unxmesh | 1 | Normal vector up direction (mesh), x component | Boundaries 1–28 | |
| ht.unymesh | unymesh | 1 | Normal vector up direction (mesh), y component | Boundaries 1–28 | |
| ht.unzmesh | unzmesh | 1 | Normal vector up direction (mesh), z component | Boundaries 1–28 | |
| ht.dEiInt | 0 | W | Total accumulated heat rate | Global | + operation |
| ht.dEi0Int | 0 | W | Total accumulated energy rate | Global | + operation |
| ht.ntfluxInt | ht.intExtBnd(ht.ntflux*ht.varIntSpa)+ht.intIntBnd(ht.ncflux_u*up(ht.varIntSpa)+ht.ncflux_d*down(ht.varIntSpa)) | W | Total net heat rate | Global | |
| ht.ntefluxInt | ht.intExtBnd(ht.nteflux*ht.varIntSpa)+ht.intIntBnd(ht.nthflux_u*up(ht.varIntSpa)+ht.nthflux_d*down(ht.varIntSpa)) | W | Total net energy rate | Global | |
| ht.QInt | ht.intDom(ht.Qtot*ht.varIntSpa)+ht.intIntLine(ht.Qltot*ht.varIntSpa)+ht.intLine(ht.Qlrtot*ht.varIntSpa)+ht.intIntPnt(ht.Qptot)+ht.intPnt(ht.Qprtot)-ht.intIntBnd(ht.ndflux_u*up(ht.varIntSpa)+ht.ndflux_d*down(ht.varIntSpa)) | W | Total heat source | Global | |
| ht.id | 1 | 1 | Physics indicator | Domains 1–4 | |
| ht.Wstr | 0 | W/m³ | Total stress power | Domains 1–4 | + operation |
| ht.WstrInt | 0 | W | Total stress power | Global | + operation |
| ht.Wtot | 0 | W/m³ | Total work source | Domains 1–4 | + operation |
| ht.WBndTot_u | 0 | W/m² | Total work source, upside | Boundaries 2–28 | + operation |
| ht.WBndTot_d | 0 | W/m² | Total work source, downside | Boundaries 1–22 | + operation |
| ht.WInt | 0 | W | Total work source | Global | + operation |
| ht.varIntSpa | ht.d | 1 | Intermediate variable | Domains 1–4 | Meta |

**2.5.3. Biological Tissue 1**




*Biological Tissue 1*

| Selection | |
|---|---|
| Geometric entity level | Domain |
| Selection | Geometry geom1: Dimension 3: All domains |

Equations

$$\rho C_p \frac{\partial T}{\partial t} + \rho C_p \mathbf{u} \cdot \nabla T + \nabla \cdot \mathbf{q} = Q + Q_{\mathrm{bio}}$$

$$\mathbf{q} = -k\nabla T$$

**Heat Conduction, Solid**

Settings

| Description | Value |
|---|---|
| Thermal conductivity | From material |

**Thermodynamics, Solid**

Settings

| Description | Value |
|---|---|
| Density | From material |
| Heat capacity at constant pressure | From material |

**Coordinate System Selection**

Settings

| Description | Value |
|---|---|
| Coordinate system | Global coordinate system |

Properties from material

| Property | Material | Property group |
|---|---|---|
| Thermal conductivity | Liver (human) | Basic |
| Density | Liver (human) | Basic |
| Heat capacity at constant pressure | Liver (human) | Basic |

**Variables**

| Name | Expression | Unit | Description | Selection | Details |
|---|---|---|---|---|---|
| ht.ndflux_u | -uflux_spatial(T)/up(ht.varIntSpa) | W/m² | Internal normal conductive heat flux, upside | Boundaries 2–15, 17, 19–28 | + operation |
| ht.ndflux_d | -dflux_spatial(T)/down(ht.varIntSpa) | W/m² | Internal normal conductive heat flux, downside | Boundary 1 | + operation |
| domflux.Tx | ht.dfluxx*ht.d | W/m² | Domain flux, x component | Domain 1 | |
| domflux.Ty | ht.dfluxy*ht.d | W/m² | Domain flux, y component | Domain 1 | |

| | | | | | |
|---|---|---|---|---|---|
| domflux.Tz | ht.dfluxz*ht.d | W/m² | Domain flux, z component | Domain 1 | |
| ht.alphap | -d(ht.rho,T)/max(ht.rho,eps) | 1/K | Isobaric compressibility coefficient | Domain 1 | |
| ht.DeltaH | integrate(subst(ht.Cp,ht.bt1.minput_pressure,ht.pref),T,ht.DeltaH_Tlow,T)+integrate (ht.mujtT,ht.bt1.minput_pressure,ht.DeltaH_plow,ht.pA) | J/kg | Sensible enthalpy | Domain 1 | + operation |
| ht.H | ht.HRef+ht.DeltaH | J/kg | Enthalpy | Domain 1 | + operation |
| ht.Ei | ht.H | J/kg | Internal energy | Domain 1 | + operation |
| ht.Ek | 0.5*(ht.ux^2+ht.uy^2+ht.uz^2) | J/kg | Kinetic energy | Domain 1 | + operation |
| ht.dfluxx | -ht.k_effxx*Tx-ht.k_effxy*Ty-ht.k_effxz*Tz | W/m² | Conductive heat flux, x component | Domain 1 | + operation |
| ht.dfluxy | -ht.k_effyx*Tx-ht.k_effyy*Ty-ht.k_effyz*Tz | W/m² | Conductive heat flux, y component | Domain 1 | + operation |
| ht.dfluxz | -ht.k_effzx*Tx-ht.k_effzy*Ty-ht.k_effzz*Tz | W/m² | Conductive heat flux, z component | Domain 1 | + operation |
| ht.dfluxtestx | -ht.k_effxx*test(Tx)-ht.k_effxy*test(Ty)-ht.k_effxz*test(Tz) | W/m² | Conductive heat flux, x component | Domain 1 | + operation |
| ht.dfluxtesty | -ht.k_effyx*test(Tx)-ht.k_effyy*test(Ty)-ht.k_effyz*test(Tz) | W/m² | Conductive heat flux, y component | Domain 1 | + operation |
| ht.dfluxtestz | -ht.k_effzx*test(Tx)-ht.k_effzy*test(Ty)-ht.k_effzz*test(Tz) | W/m² | Conductive heat flux, z component | Domain 1 | + operation |
| ht.cfluxx | ht.rho*ht.ux*ht.Ei | W/m² | Convective heat flux, x component | Domain 1 | + operation |
| ht.cfluxy | ht.rho*ht.uy*ht.Ei | W/m² | Convective heat flux, y component | Domain 1 | + operation |
| ht.cfluxz | ht.rho*ht.uz*ht.Ei | W/m² | Convective heat flux, z component | Domain 1 | + operation |
| ht.tefluxx | ht.dfluxx+ht.thfluxx | W/m² | Total energy flux, x component | Domain 1 | + operation |
| ht.tefluxy | ht.dfluxy+ht.thfluxy | W/m² | Total energy flux, y component | Domain 1 | + operation |
| ht.tefluxz | ht.dfluxz+ht.thfluxz | W/m² | Total energy flux, z component | Domain 1 | + operation |
| ht.thfluxx | ht.rho*ht.ux*ht.H0 | W/m² | Total enthalpy flux, x component | Domain 1 | + operation |
| ht.thfluxy | ht.rho*ht.uy*ht.H0 | W/m² | Total enthalpy flux, y component | Domain 1 | + operation |
| ht.thfluxz | ht.rho*ht.uz*ht.H0 | W/m² | Total enthalpy flux, z component | Domain 1 | + operation |
| ht.dEiInt | ht.bt1.dEiInt | W | Total accumulated heat rate | Global | + operation |
| ht.dEi0Int | ht.bt1.dEi0Int | W | Total accumulated energy rate | Global | + operation |
| ht.Wstr | ht.pA*(d(ht.ux,x)+d(ht.uy,y)+d(ht.uz,z)) | W/m³ | Total stress power | Domain 1 | + operation |
| ht.WstrInt | ht.bt1.WstrInt | W | Total stress power | Global | + operation |
| ht.WInt | ht.bt1.WInt | W | Total work source | Global | + operation |
| ht.kxx | material.k11 | W/(m·K) | Thermal conductivity, xx component | Domain 1 | Meta |
| ht.kyx | material.k21 | W/(m·K) | Thermal conductivity, yx component | Domain 1 | Meta |
| ht.kzx | material.k31 | W/(m·K) | Thermal conductivity, zx component | Domain 1 | Meta |
| ht.kxy | material.k12 | W/(m·K) | Thermal conductivity, xy component | Domain 1 | Meta |
| ht.kyy | material.k22 | W/(m·K) | Thermal conductivity, yy component | Domain 1 | Meta |
| ht.kzy | material.k32 | W/(m·K) | Thermal conductivity, zy component | Domain 1 | Meta |
| ht.kxz | material.k13 | W/(m·K) | Thermal conductivity, xz component | Domain 1 | Meta |
| ht.kyz | material.k23 | W/(m·K) | Thermal conductivity, yz component | Domain 1 | Meta |
| ht.kzz | material.k33 | W/(m·K) | | Domain 1 | Meta |

| | | | | | |
|---|---|---|---|---|---|
| | | | Thermal conductivity, zz component | | |
| ht.k_iso | material.k_iso | W/(m·K) | Thermal conductivity, isotropic value | Domain 1 | Meta |
| ht.rho | material.rho | kg/m³ | Density | Domain 1 | Meta |
| ht.Cp | material.Cp | J/(kg·K) | Heat capacity at constant pressure | Domain 1 | Meta |
| ht.res_T | ht.timeDerivative*ht.C_eff-ht.k_effxx*Txx-ht.k_effxy*Txy-ht.k_effxz*Txz-ht.k_effyx*Tyx-ht.k_effyy*Tyy-ht.k_effyz*Tyz-ht.k_effzx*Tzx-ht.k_effzy*Tzy-ht.k_effzz*Tzz-(ht.qs+ht.qs_oop)*T+ht.C_eff*(ht.ux*Tx+ht.uy*Ty+ht.uz*Tz)-ht.D_Hx*Tx-ht.D_Hy*Ty-ht.D_Hz*Tz+ht.rholCplUlx*Tx+ht.rholCplUly*Ty+ht.rholCplUlz*Tz-ht.Q-ht.Qoop | W/m³ | Equation residual | Domain 1 | + operation |
| ht.pA | ht.pref | Pa | Absolute pressure | Domain 1 | |
| ht.Qmet | 0 | W/m³ | Metabolic heat source | Domain 1 | + operation |
| ht.pref | ht.bt1.pref | Pa | Reference pressure level | Domain 1 | |
| ht.DeltaH_Tlow | ht.Tref | K | Temperature lower bound for enthalpy evaluation | Domain 1 | |
| ht.DeltaH_plow | ht.pref | Pa | Pressure lower bound for enthalpy evaluation | Domain 1 | |
| ht.mujtT | 0 | m³/kg | Isothermal Joule-Thomson coefficient | Domain 1 | |
| ht.alphap_ref | -d(root.comp1.ht.rhoref,ht.Tref)/max(root.comp1.ht.rhoref,eps) | 1/K | Reference isobaric compressibility coefficient | Domain 1 | |
| ht.alphapT | ht.alphap*T | 1 | Help variable | Domain 1 | |
| ht.rhoInit | subst(ht.rho,T,ht.Tinit,minput.pA,ht.pref) | kg/m³ | Initial density | Domain 1 | |
| ht.rho_eff | ht.rho | kg/m³ | Effective density | Domain 1 | |
| ht.C_eff | ht.rho*ht.Cp | J/(m³·K) | Effective volumetric heat capacity | Domain 1 | |
| ht.k_effxx | ht.kxx | W/(m·K) | Effective thermal conductivity, xx component | Domain 1 | |
| ht.k_effyx | ht.kyx | W/(m·K) | Effective thermal conductivity, yx component | Domain 1 | |
| ht.k_effzx | ht.kzx | W/(m·K) | Effective thermal conductivity, zx component | Domain 1 | |
| ht.k_effxy | ht.kxy | W/(m·K) | Effective thermal conductivity, xy component | Domain 1 | |
| ht.k_effyy | ht.kyy | W/(m·K) | Effective thermal conductivity, yy component | Domain 1 | |
| ht.k_effzy | ht.kzy | W/(m·K) | Effective thermal conductivity, zy component | Domain 1 | |
| ht.k_effxz | ht.kxz | W/(m·K) | Effective thermal conductivity, xz component | Domain 1 | |
| ht.k_effyz | ht.kyz | W/(m·K) | Effective thermal conductivity, yz component | Domain 1 | |
| ht.k_effzz | ht.kzz | W/(m·K) | Effective thermal conductivity, zz component | Domain 1 | |
| ht.kappaTxx | 0 | W/(m·K) | Turbulent thermal conductivity, xx component | Domain 1 | |
| ht.kappaTyx | 0 | W/(m·K) | Turbulent thermal conductivity, yx component | Domain 1 | |
| ht.kappaTzx | 0 | W/(m·K) | Turbulent thermal conductivity, zx component | Domain 1 | |
| ht.kappaTxy | 0 | | | Domain 1 | |

| | | | | | |
|---|---|---|---|---|---|
| | | W/(m·K) | Turbulent thermal conductivity, xy component | | |
| ht.kappaTyy | 0 | W/(m·K) | Turbulent thermal conductivity, yy component | Domain 1 | |
| ht.kappaTzy | 0 | W/(m·K) | Turbulent thermal conductivity, zy component | Domain 1 | |
| ht.kappaTxz | 0 | W/(m·K) | Turbulent thermal conductivity, xz component | Domain 1 | |
| ht.kappaTyz | 0 | W/(m·K) | Turbulent thermal conductivity, yz component | Domain 1 | |
| ht.kappaTzz | 0 | W/(m·K) | Turbulent thermal conductivity, zz component | Domain 1 | |
| ht.kmean | (ht.k_effxx+ht.k_effyy+ht.k_effzz)/3 | W/(m·K) | Mean effective thermal conductivity | Domain 1 | |
| ht.ux | 0 | m/s | Velocity field, x component | Domain 1 | + operation |
| ht.uy | 0 | m/s | Velocity field, y component | Domain 1 | + operation |
| ht.uz | 0 | m/s | Velocity field, z component | Domain 1 | + operation |
| ht.cellPe | 0.5*ht.rho*ht.Cp*h*sqrt(ht.ux^2+ht.uy^2+ht.uz^2)/ht.kmean | 1 | Cell Péclet number | Domain 1 | |
| ht.gradTx | Tx | K/m | Temperature gradient, x component | Domain 1 | |
| ht.gradTy | Ty | K/m | Temperature gradient, y component | Domain 1 | |
| ht.gradTz | Tz | K/m | Temperature gradient, z component | Domain 1 | |
| ht.gradTmag | sqrt(ht.gradTx^2+ht.gradTy^2+ht.gradTz^2) | K/m | Temperature gradient magnitude | Domain 1 | |
| ht.alphaTdxx | ht.k_effxx/ht.C_eff | m²/s | Thermal diffusivity, xx component | Domain 1 | |
| ht.alphaTdyx | ht.k_effyx/ht.C_eff | m²/s | Thermal diffusivity, yx component | Domain 1 | |
| ht.alphaTdzx | ht.k_effzx/ht.C_eff | m²/s | Thermal diffusivity, zx component | Domain 1 | |
| ht.alphaTdxy | ht.k_effxy/ht.C_eff | m²/s | Thermal diffusivity, xy component | Domain 1 | |
| ht.alphaTdyy | ht.k_effyy/ht.C_eff | m²/s | Thermal diffusivity, yy component | Domain 1 | |
| ht.alphaTdzy | ht.k_effzy/ht.C_eff | m²/s | Thermal diffusivity, zy component | Domain 1 | |
| ht.alphaTdxz | ht.k_effxz/ht.C_eff | m²/s | Thermal diffusivity, xz component | Domain 1 | |
| ht.alphaTdyz | ht.k_effyz/ht.C_eff | m²/s | Thermal diffusivity, yz component | Domain 1 | |
| ht.alphaTdzz | ht.k_effzz/ht.C_eff | m²/s | Thermal diffusivity, zz component | Domain 1 | |
| ht.alphaTdMean | ht.kmean/ht.C_eff | m²/s | Mean thermal diffusivity | Domain 1 | |
| ht.Tradu | ht.Tu | K | Upside temperature | Domain 1 | |
| ht.Tradu | ht.Tu | K | Upside temperature | Boundaries 1–15, 17, 19–28 | |
| ht.Tradd | ht.Td | K | Downside temperature | Domain 1 | |
| | | | | | |

| | | | | | |
|---|---|---|---|---|---|
| ht.Tradd | ht.Td | K | Downside temperature | Boundaries 1–15, 17, 19–28 | |
| ht.dEi | material.dt(ht.rho*ht.Ei) | W/m³ | Total accumulated heat rate density | Domain 1 | |
| ht.dEi0 | material.dt(ht.rho*ht.Ei0) | W/m³ | Total accumulated energy rate density | Domain 1 | |
| ht.D_Hx | 0 | W/(m²·K) | Enthalpy diffusion coefficient, x component | Domain 1 | |
| ht.D_Hy | 0 | W/(m²·K) | Enthalpy diffusion coefficient, y component | Domain 1 | |
| ht.D_Hz | 0 | W/(m²·K) | Enthalpy diffusion coefficient, z component | Domain 1 | |
| ht.Q_H | ht.D_Hx*Tx+ht.D_Hy*Ty+ht.D_Hz*Tz | W/m³ | Enthalpy diffusion flux | Domain 1 | |
| ht.rholCplUlx | 0 | W/(m²·K) | Heat transfer coefficient, liquid water transport, x component | Domain 1 | |
| ht.rholCplUly | 0 | W/(m²·K) | Heat transfer coefficient, liquid water transport, y component | Domain 1 | |
| ht.rholCplUlz | 0 | W/(m²·K) | Heat transfer coefficient, liquid water transport, z component | Domain 1 | |
| ht.timeDerivative | material.dt(T) | K/s | Temperature, first time derivative | Domain 1 | |
| ht.gamma | 1 | 1 | Ratio of specific heats | Domain 1 | |
| ht.Trho | ht.Tref | K | Temperature for density evaluation | Domain 1 | |
| ht.dfltopaque | 1 | 1 | Default opacity | Domain 1 | |
| ht.helem | h_spatial | m | Element size | Domain 1 | |
| ht.bt1.pref | 1[atm] | Pa | Reference pressure level | Domain 1 | |
| ht.bt1.dEiInt | ht.bt1.intDom((ht.dEi-ht.Qm*ht.Ei)*ht.varIntSpa) | W | Total accumulated heat rate | Global | |
| ht.bt1.dEi0Int | ht.bt1.intDom((ht.dEi0-ht.Qm*ht.H)*ht.varIntSpa) | W | Total accumulated energy rate | Global | |
| ht.bt1.ntfluxInt | ht.bt1.intExtBnd(ht.ntflux*ht.varIntSpa)+ht.bt1.intExtBndUp(ht.ntflux_u*up(ht.varIntSpa))+ht.bt1.intExtBndDown(ht.ntflux_d*down(ht.varIntSpa))+ht.bt1.intIntBnd(ht.ncflux_u*up(ht.varIntSpa)+ht.ncflux_d*down(ht.varIntSpa)) | W | Total net heat rate | Global | |
| ht.bt1.ntefluxInt | ht.bt1.intExtBnd(ht.nteflux*ht.varIntSpa)+ht.bt1.intExtBndUp(ht.nteflux_u*up(ht.varIntSpa))+ht.bt1.intExtBndDown(ht.nteflux_d*down(ht.varIntSpa))+ht.bt1.intIntBnd(ht.nthflux_u*up(ht.varIntSpa)+ht.nthflux_d*down(ht.varIntSpa)) | W | Total net energy rate | Global | |
| ht.bt1.QInt | ht.bt1.intDom(ht.Qtot*ht.varIntSpa)+ht.bt1.intIntLine(ht.Qltot*ht.varIntSpa)+ht.intLine(subst(ht.Qlrtot,ht.id,isdefined(ht.bt1.id))*ht.varIntSpa)+ht.bt1.intIntPnt(ht.Qptot)+ht.intPnt(subst(ht.Qprtot,ht.id,isdefined(ht.bt1.id)))-ht.bt1.intIntBnd(ht.ndflux_u*up(ht.varIntSpa)+ht.ndflux_d*down(ht.varIntSpa)) | W | Total heat source | Global | |
| ht.bt1.WstrInt | ht.bt1.intDom(ht.Wstr*ht.varIntSpa) | W | Total stress power | Global | |
| ht.bt1.WInt | ht.bt1.intDom(ht.Wtot*ht.varIntSpa)+ht.bt1.intBndUp(ht.WBndTot_u*up(ht.varIntSpa))+ht.bt1.intBndDown(ht.WBndTot_d*down(ht.varIntSpa)) | W | Total work source | Global | |

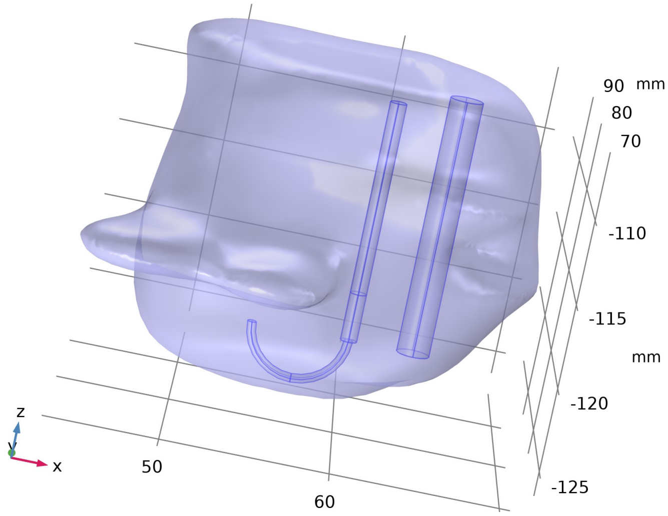
#### Shape functions

| Name | Shape function | Unit | Description | Shape frame | Selection |
|---|---|---|---|---|---|
| T | Lagrange (Quadratic) | K | Temperature | Spatial | Domain 1 |

#### Weak Expressions

| Weak expression | Integration order | Integration frame | Selection |
|---|---|---|---|
| ht.streamline | 4 | Spatial | Domain 1 |
| (ht.dfluxx*test(Tx)+ht.dfluxy*test(Ty)+ht.dfluxz*test(Tz))*ht.d | 4 | Spatial | Domain 1 |
| -ht.C_eff*ht.timeDerivative*test(T)*ht.d | 4 | Spatial | Domain 1 |
| -ht.C_eff*(ht.ux*Tx+ht.uy*Ty+ht.uz*Tz)*test(T)*ht.d | 4 | Spatial | Domain 1 |

#### Bioheat 1




*Bioheat 1*

Selection

| | |
|---|---|
| Geometric entity level | Domain |
| Selection | Geometry geom1: Dimension 3: All domains |

Equations

$$\rho C_p \frac{\partial T}{\partial t} + \rho C_p \mathbf{u} \cdot \nabla T + \nabla \cdot \mathbf{q} = Q + Q_{bio}$$

$$Q_{bio} = \rho_b C_{p,b} \omega_b (T_b - T) + Q_{met}$$

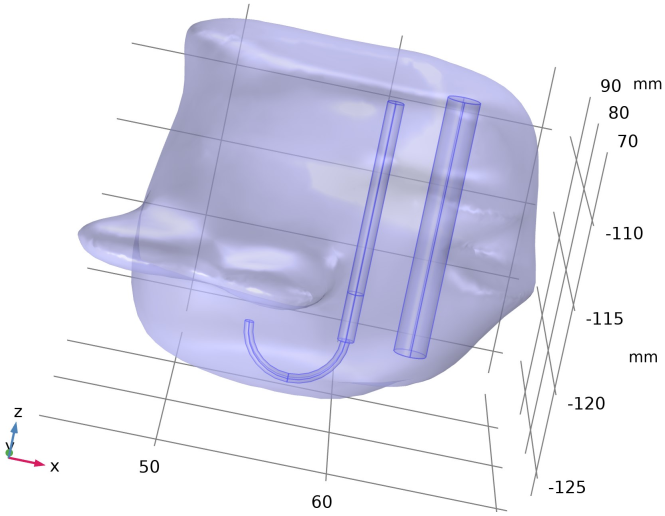
**Variables**

| Name | Expression | Unit | Description | Selection | Details |
|---|---|---|---|---|---|
| ht.Q | ht.rhobl*ht.Cp_b*ht.omegab*(ht.Tb-T)+ht.Qmet | W/m³ | Heat source | Domain 1 | + operation |
| ht.Qtot | ht.rhobl*ht.Cp_b*ht.omegab*(ht.Tb-T)+ht.Qmet | W/m³ | Total heat source | Domain 1 | + operation |
| ht.Qmet | 0 | W/m³ | Metabolic heat source | Domain 1 | + operation |
| ht.Tb | T_b | K | Arterial blood temperature | Domain 1 | |
| ht.Cp_b | c_b | J/(kg·K) | Specific heat, blood | Domain 1 | |
| ht.omegab | omega_b | 1/s | Blood perfusion rate | Domain 1 | |
| ht.rhobl | rho_b*spatial.detInvF | kg/m³ | Density, blood | Domain 1 | |

**Weak Expressions**

| Weak expression | Integration order | Integration frame | Selection |
|---|---|---|---|
| ht.rhobl*ht.Cp_b*ht.omegab*(ht.Tb-T)*test(T)*ht.d | 4 | Spatial | Domain 1 |
| ht.Qmet*test(T)*ht.d | 4 | Spatial | Domain 1 |

**Thermal Damage 1**




*Thermal Damage 1*

Selection

| | |
|---|---|
| Geometric entity level | Domain |
| Selection | Geometry geom1: Dimension 3: All domains |

Equations

$$\rho C_p \frac{\partial T}{\partial t} + \rho C_p \mathbf{u} \cdot \nabla T + \nabla \cdot \mathbf{q} = Q + Q_{\text{bio}}$$

$$\mathbf{q} = -k \nabla T$$

$$Q = -\rho L \frac{\partial \theta_{\text{d}}}{\partial t}$$

$$\theta_{\text{d}} = \alpha$$

$$\frac{\partial \alpha}{\partial t} = (1-\alpha)^n A e^{\frac{-\Delta E}{RT}}$$

##### Damaged Tissue

Settings

| Description | Value |
|---|---|
| Transformation model | Arrhenius kinetics |
| Frequency factor | From material |
| Activation energy | From material |
| Polynomial order | 1 |
| Enthalpy change | 0 |
| Specify different material properties for damaged tissue | Off |

##### Initial Values

Settings

| Description | Value |
|---|---|
| Initial damaged tissue indicator | 0 |

Properties from material

| Property | Material | Property group |
|---|---|---|
| Frequency factor | Liver (human) | Basic |
| Activation energy | Liver (human) | Basic |

##### Variables

| Name | Expression | Unit | Description | Selection | Details |
|---|---|---|---|---|---|
| ht.Qtot | ht.Qdmg | W/m³ | Total heat source | Domain 1 | + operation |
| ht.L | 0 | J/kg | Enthalpy change | Domain 1 | |
| ht.bt1.tdam1.n | 1 | 1 | Polynomial order | Domain 1 | |
| ht.A | material.A | 1/s | Frequency factor | Domain 1 | Meta |

| | | | | | |
|---|---|---|---|---|---|
| ht.deltaE | material.deltaE | J/mol | Activation energy | Domain 1 | Meta |
| ht.theta_d | max(0,min(ht.alpha,1)) | 1 | Fraction of damage | Domain 1 | |
| ht.bt1.tdam1.theta_dt | (1-ht.alpha)^ht.bt1.tdam1.n*ht.A*exp(-ht.deltaE/(R_const*T)) | 1/s | Rate of damage | Domain 1 | |
| ht.bt1.tdam1.L | ht.L | J/kg | Enthalpy change | Domain 1 | |
| ht.alpha_init | 0 | 1 | Initial damaged tissue indicator | Domain 1 | |
| ht.bt1.tdam1.sigma | 1 | | Penalty factor | Domain 1 | |
| ht.Qdmg | -ht.rho*ht.bt1.tdam1.L*ht.bt1.tdam1.theta_dt | W/m³ | Damage heat source | Domain 1 | |
| ht.theta_d_sm | max(0,min(1,ht.theta_d_post)) | 1 | Fraction of necrotic tissue, smoothing | Domain 1 | |
| ht.theta_d_post | ht.bt1.tdam1.theta_d_post | 1 | Fraction of damage | Domain 1 | |

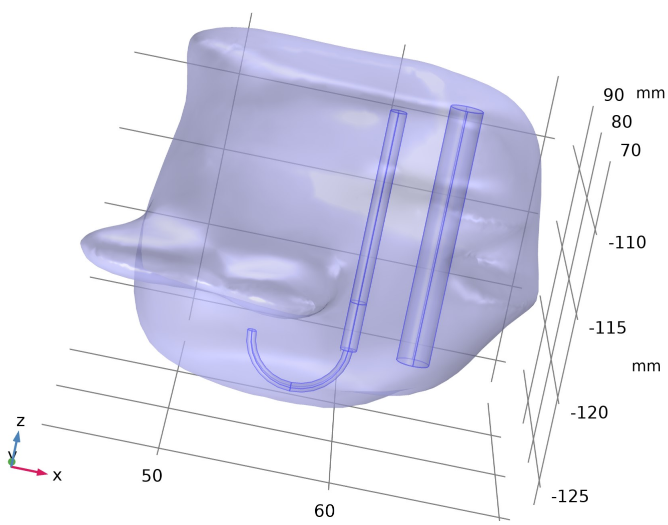
**Shape functions**

| Name | Shape function | Unit | Description | Shape frame | Selection |
|---|---|---|---|---|---|
| ht.alpha | Discontinuous Lagrange (Constant) | | Degree of tissue injury | Spatial | Domain 1 |

**Weak Expressions**

| Weak expression | Integration order | Integration frame | Selection |
|---|---|---|---|
| (ht.bt1.tdam1.sigma*(1-ht.alpha)^ht.bt1.tdam1.n*ht.A*exp(-ht.deltaE/(R_const*T))-ht.alphat)*test(ht.alpha) | 4 | Spatial | Domain 1 |
| ht.Qdmg*test(T)*ht.d | 4 | Spatial | Domain 1 |

#### 2.5.4. Initial Values 1




*Initial Values 1*

Selection

| | |
|---|---|
| Geometric entity level | Domain |
| Selection | Geometry geom1: Dimension 3: All domains |

**Initial Values**

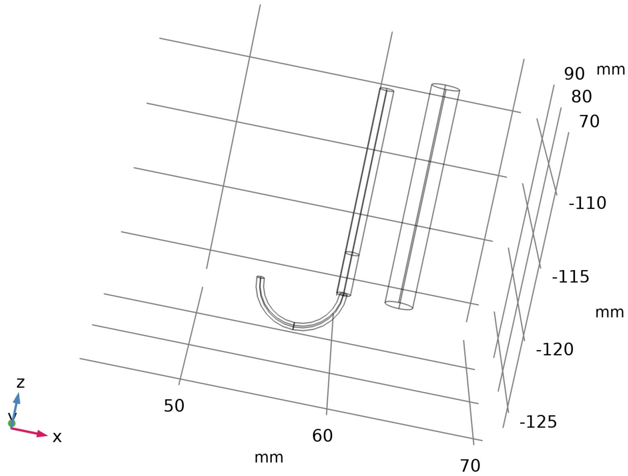
Settings

| Description | Value |
|---|---|
| Temperature | User defined |
| Temperature | T0 |

**Variables**

| Name | Expression | Unit | Description | Selection |
|---|---|---|---|---|
| ht.Tinit | T0 | K | Temperature | Domains 1–4 |

#### 2.5.5. Thermal Insulation 1




*Thermal Insulation 1*

Selection

| | |
|---|---|
| Geometric entity level | Boundary |
| Selection | Geometry geom1: Dimension 2: All boundaries |

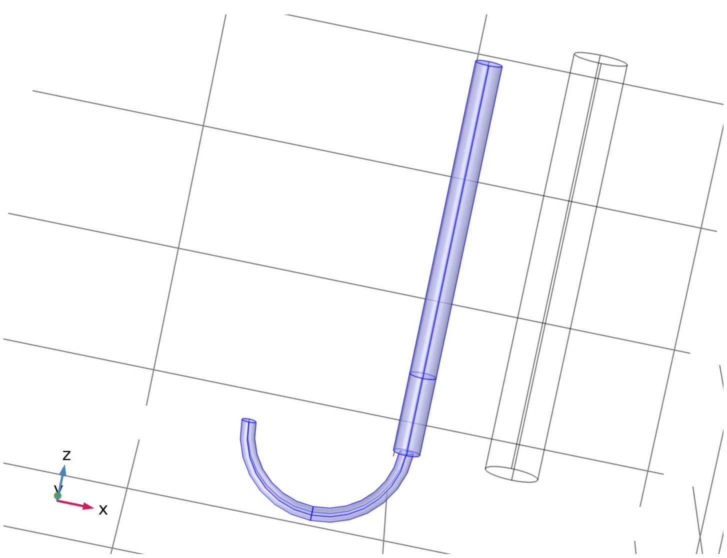
Equations

$$-\mathbf{n} \cdot \mathbf{q} = 0$$

**Shape functions**

| Name | Shape function | Unit | Description | Shape frame | Selection | Details |
|---|---|---|---|---|---|---|
| T | Lagrange (Quadratic) | K | Temperature | Spatial | No boundaries | Slit |
| T | Lagrange (Quadratic) | K | Temperature | Material | No boundaries | Slit |
| T | Lagrange (Quadratic) | K | Temperature | Geometry | No boundaries | Slit |
| T | Lagrange (Quadratic) | K | Temperature | Mesh | No boundaries | Slit |

**2.5.6. Solid 1**




*Solid 1*

Selection

| | |
|---|---|
| Geometric entity level | Domain |
| Name | Trocar |
| Selection | Named uni1: Geometry geom1: Dimension 3: Domains 2–4 |

Equations

$$\rho C_p \frac{\partial T}{\partial t} + \rho C_p \mathbf{u} \cdot \nabla T + \nabla \cdot \mathbf{q} = Q + Q_{ted}$$

$$\mathbf{q} = -k \nabla T$$

#### Heat Conduction, Solid

Settings

| Description | Value |
|---|---|
| Thermal conductivity | From material |

#### Thermodynamics, Solid

Settings

| Description | Value |
|---|---|
| Density | From material |
| Heat capacity at constant pressure | From material |

#### Coordinate System Selection

Settings

| Description | Value |
|---|---|
| Coordinate system | Global coordinate system |

Properties from material

| Property | Material | Property group |
|---|---|---|
| Thermal conductivity | Electrodes | Basic |
| Density | Electrodes | Basic |
| Heat capacity at constant pressure | Electrodes | Basic |
| Thermal conductivity | Trocar Tip | Basic |
| Density | Trocar Tip | Basic |
| Heat capacity at constant pressure | Trocar Tip | Basic |
| Thermal conductivity | Trocar Base | Basic |
| Density | Trocar Base | Basic |
| Heat capacity at constant pressure | Trocar Base | Basic |

#### Variables

| Name | Expression | Unit | Description | Selection | Details |
|---|---|---|---|---|---|
| ht.ndflux_u | -uflux_spatial(T)/up(ht.varIntSpa) | W/m² | Internal normal conductive heat flux, upside | Boundaries 16, 18 | + operation |
| ht.ndflux_d | -dflux_spatial(T)/down(ht.varIntSpa) | W/m² | Internal normal conductive heat flux, downside | Boundaries 16, 18 | + operation |
| ht.ndflux_d | -dflux_spatial(T)/down(ht.varIntSpa) | W/m² | Internal normal conductive heat flux, downside | Boundaries 2–15, 17, 19–22 | + operation |
| domflux.Tx | ht.dfluxx*ht.d | W/m² | Domain flux, x component | Domains 2–4 | |
| domflux.Ty | ht.dfluxy*ht.d | W/m² | Domain flux, y component | Domains 2–4 | |
| domflux.Tz | ht.dfluxz*ht.d | W/m² | Domain flux, z component | Domains 2–4 | |
| ht.alphap | -d(ht.rho,T)/max(ht.rho,eps) | 1/K | Isobaric compressibility coefficient | Domains 2–4 | |
| ht.DeltaH | integrate(subst(ht.Cp,ht.solid1.minput_pressure,ht.pref),T,ht.DeltaH_Tlow,T)+integrate(ht.mujtT,ht.solid1.minput_pressure,ht.DeltaH_plow,ht.pA) | J/kg | Sensible enthalpy | Domains 2–4 | + operation |
| ht.H | ht.HRef+ht.DeltaH | J/kg | Enthalpy | Domains 2–4 | + operation |
| ht.Ei | ht.H | J/kg | Internal energy | Domains 2–4 | + operation |
| ht.Ek | 0.5*(ht.ux^2+ht.uy^2+ht.uz^2) | J/kg | Kinetic energy | Domains 2–4 | + operation |
| ht.dfluxx | -ht.k_effxx*Tx-ht.k_effxy*Ty-ht.k_effxz*Tz | W/m² | Conductive heat flux, x component | Domains 2–4 | + operation |
| ht.dfluxy | -ht.k_effyx*Tx-ht.k_effyy*Ty-ht.k_effyz*Tz | W/m² | Conductive heat flux, y component | Domains 2–4 | + operation |
| ht.dfluxz | -ht.k_effzx*Tx-ht.k_effzy*Ty-ht.k_effzz*Tz | W/m² | Conductive heat flux, z component | Domains 2–4 | + operation |
| ht.dfluxtestx | -ht.k_effxx*test(Tx)-ht.k_effxy*test(Ty)-ht.k_effxz*test(Tz) | W/m² | | | + operation |

| | | | | | |
|---|---|---|---|---|---|
| | | | Conductive heat flux, x component | Domains 2–4 | |
| ht.dfluxtesty | -ht.k_effyx*test(Tx)-ht.k_effyy*test(Ty)-ht.k_effyz*test(Tz) | W/m² | Conductive heat flux, y component | Domains 2–4 | + operation |
| ht.dfluxtestz | -ht.k_effzx*test(Tx)-ht.k_effzy*test(Ty)-ht.k_effzz*test(Tz) | W/m² | Conductive heat flux, z component | Domains 2–4 | + operation |
| ht.cfluxx | ht.rho*ht.ux*ht.Ei | W/m² | Convective heat flux, x component | Domains 2–4 | + operation |
| ht.cfluxy | ht.rho*ht.uy*ht.Ei | W/m² | Convective heat flux, y component | Domains 2–4 | + operation |
| ht.cfluxz | ht.rho*ht.uz*ht.Ei | W/m² | Convective heat flux, z component | Domains 2–4 | + operation |
| ht.tefluxx | ht.dfluxx+ht.thfluxx | W/m² | Total energy flux, x component | Domains 2–4 | + operation |
| ht.tefluxy | ht.dfluxy+ht.thfluxy | W/m² | Total energy flux, y component | Domains 2–4 | + operation |
| ht.tefluxz | ht.dfluxz+ht.thfluxz | W/m² | Total energy flux, z component | Domains 2–4 | + operation |
| ht.thfluxx | ht.rho*ht.ux*ht.H0 | W/m² | Total enthalpy flux, x component | Domains 2–4 | + operation |
| ht.thfluxy | ht.rho*ht.uy*ht.H0 | W/m² | Total enthalpy flux, y component | Domains 2–4 | + operation |
| ht.thfluxz | ht.rho*ht.uz*ht.H0 | W/m² | Total enthalpy flux, z component | Domains 2–4 | + operation |
| ht.dEiInt | ht.solid1.dEiInt | W | Total accumulated heat rate | Global | + operation |
| ht.dEi0Int | ht.solid1.dEi0Int | W | Total accumulated energy rate | Global | + operation |
| ht.Wstr | ht.pA*(d(ht.ux,x)+d(ht.uy,y)+d(ht.uz,z)) | W/m³ | Total stress power | Domains 2–4 | + operation |
| ht.WstrInt | ht.solid1.WstrInt | W | Total stress power | Global | + operation |
| ht.WInt | ht.solid1.WInt | W | Total work source | Global | + operation |
| ht.kxx | material.k11 | W/(m·K) | Thermal conductivity, xx component | Domains 2–4 | Meta |
| ht.kyx | material.k21 | W/(m·K) | Thermal conductivity, yx component | Domains 2–4 | Meta |
| ht.kzx | material.k31 | W/(m·K) | Thermal conductivity, zx component | Domains 2–4 | Meta |
| ht.kxy | material.k12 | W/(m·K) | Thermal conductivity, xy component | Domains 2–4 | Meta |
| ht.kyy | material.k22 | W/(m·K) | Thermal conductivity, yy component | Domains 2–4 | Meta |
| ht.kzy | material.k32 | W/(m·K) | Thermal conductivity, zy component | Domains 2–4 | Meta |
| ht.kxz | material.k13 | W/(m·K) | Thermal conductivity, xz component | Domains 2–4 | Meta |
| ht.kyz | material.k23 | W/(m·K) | Thermal conductivity, yz component | Domains 2–4 | Meta |
| ht.kzz | material.k33 | W/(m·K) | Thermal conductivity, zz component | Domains 2–4 | Meta |
| ht.k_iso | material.k_iso | W/(m·K) | Thermal conductivity, isotropic value | Domains 2–4 | Meta |
| ht.rho | material.rho | kg/m³ | Density | Domains 2–4 | Meta |
| ht.Cp | material.Cp | J/(kg·K) | Heat capacity at constant pressure | Domains 2–4 | Meta |
| ht.res_T | ht.timeDerivative*ht.C_eff-ht.k_effxx*Txx-ht.k_effxy*Txy-ht.k_effxz*Txz-ht.k_effyx*Tyx-ht.k_effyy*Tyy-ht.k_effyz*Tyz-ht.k_effzx*Tzx-ht.k_effzy*Tzy-ht.k_effzz*Tzz-(ht.qs+ht.qs_oop)*T+ht.C_eff*(ht.ux*Tx+ht.uy*Ty+ht.uz*Tz)-ht.D_Hx*Tx-ht.D_Hy*Ty-ht.D_Hz*Tz+ht.rholCplUlx*Tx+ht.rholCplUly*Ty+ht.rholCplUlz*Tz-ht.Q-ht.Qoop | W/m³ | Equation residual | Domains 2–4 | + operation |
| ht.pA | ht.pref | Pa | Absolute pressure | Domains 2–4 | |
| | | | | | |

| | | | | | |
|---|---|---|---|---|---|
| ht.Qmet | 0 | W/m³ | Metabolic heat source | Domains 2–4 | + operation |
| ht.pref | ht.solid1.pref | Pa | Reference pressure level | Domains 2–4 | |
| ht.DeltaH_Tlow | ht.Tref | K | Temperature lower bound for enthalpy evaluation | Domains 2–4 | |
| ht.DeltaH_plow | ht.pref | Pa | Pressure lower bound for enthalpy evaluation | Domains 2–4 | |
| ht.mujtT | 0 | m³/kg | Isothermal Joule-Thomson coefficient | Domains 2–4 | |
| ht.alphap_ref | -d(root.comp1.ht.rhoref,ht.Tref)/max(root.comp1.ht.rhoref,eps) | 1/K | Reference isobaric compressibility coefficient | Domains 2–4 | |
| ht.alphapT | ht.alphap*T | 1 | Help variable | Domains 2–4 | |
| ht.rhoInit | subst(ht.rho,T,ht.Tinit,minput.pA,ht.pref) | kg/m³ | Initial density | Domains 2–4 | |
| ht.rho_eff | ht.rho | kg/m³ | Effective density | Domains 2–4 | |
| ht.C_eff | ht.rho*ht.Cp | J/(m³·K) | Effective volumetric heat capacity | Domains 2–4 | |
| ht.k_effxx | ht.kxx | W/(m·K) | Effective thermal conductivity, xx component | Domains 2–4 | |
| ht.k_effyx | ht.kyx | W/(m·K) | Effective thermal conductivity, yx component | Domains 2–4 | |
| ht.k_effzx | ht.kzx | W/(m·K) | Effective thermal conductivity, zx component | Domains 2–4 | |
| ht.k_effxy | ht.kxy | W/(m·K) | Effective thermal conductivity, xy component | Domains 2–4 | |
| ht.k_effyy | ht.kyy | W/(m·K) | Effective thermal conductivity, yy component | Domains 2–4 | |
| ht.k_effzy | ht.kzy | W/(m·K) | Effective thermal conductivity, zy component | Domains 2–4 | |
| ht.k_effxz | ht.kxz | W/(m·K) | Effective thermal conductivity, xz component | Domains 2–4 | |
| ht.k_effyz | ht.kyz | W/(m·K) | Effective thermal conductivity, yz component | Domains 2–4 | |
| ht.k_effzz | ht.kzz | W/(m·K) | Effective thermal conductivity, zz component | Domains 2–4 | |
| ht.kappaTxx | 0 | W/(m·K) | Turbulent thermal conductivity, xx component | Domains 2–4 | |
| ht.kappaTyx | 0 | W/(m·K) | Turbulent thermal conductivity, yx component | Domains 2–4 | |
| ht.kappaTzx | 0 | W/(m·K) | Turbulent thermal conductivity, zx component | Domains 2–4 | |
| ht.kappaTxy | 0 | W/(m·K) | Turbulent thermal conductivity, xy component | Domains 2–4 | |
| ht.kappaTyy | 0 | W/(m·K) | Turbulent thermal conductivity, yy component | Domains 2–4 | |
| ht.kappaTzy | 0 | W/(m·K) | Turbulent thermal conductivity, zy component | Domains 2–4 | |
| ht.kappaTxz | 0 | W/(m·K) | Turbulent thermal conductivity, xz component | Domains 2–4 | |
| ht.kappaTyz | 0 | W/(m·K) | | Domains 2–4 | |

| | | | | | |
|---|---|---|---|---|---|
| | | | Turbulent thermal conductivity, yz component | | |
| ht.kappaTzz | 0 | W/(m·K) | Turbulent thermal conductivity, zz component | Domains 2–4 | |
| ht.kmean | (ht.k_effxx+ht.k_effyy+ht.k_effzz)/3 | W/(m·K) | Mean effective thermal conductivity | Domains 2–4 | |
| ht.ux | 0 | m/s | Velocity field, x component | Domains 2–4 | + operation |
| ht.uy | 0 | m/s | Velocity field, y component | Domains 2–4 | + operation |
| ht.uz | 0 | m/s | Velocity field, z component | Domains 2–4 | + operation |
| ht.cellPe | 0.5*ht.rho*ht.Cp*h*sqrt(ht.ux^2+ht.uy^2+ht.uz^2)/ht.kmean | 1 | Cell Péclet number | Domains 2–4 | |
| ht.gradTx | Tx | K/m | Temperature gradient, x component | Domains 2–4 | |
| ht.gradTy | Ty | K/m | Temperature gradient, y component | Domains 2–4 | |
| ht.gradTz | Tz | K/m | Temperature gradient, z component | Domains 2–4 | |
| ht.gradTmag | sqrt(ht.gradTx^2+ht.gradTy^2+ht.gradTz^2) | K/m | Temperature gradient magnitude | Domains 2–4 | |
| ht.alphaTdxx | ht.k_effxx/ht.C_eff | m²/s | Thermal diffusivity, xx component | Domains 2–4 | |
| ht.alphaTdyx | ht.k_effyx/ht.C_eff | m²/s | Thermal diffusivity, yx component | Domains 2–4 | |
| ht.alphaTdzx | ht.k_effzx/ht.C_eff | m²/s | Thermal diffusivity, zx component | Domains 2–4 | |
| ht.alphaTdxy | ht.k_effxy/ht.C_eff | m²/s | Thermal diffusivity, xy component | Domains 2–4 | |
| ht.alphaTdyy | ht.k_effyy/ht.C_eff | m²/s | Thermal diffusivity, yy component | Domains 2–4 | |
| ht.alphaTdzy | ht.k_effzy/ht.C_eff | m²/s | Thermal diffusivity, zy component | Domains 2–4 | |
| ht.alphaTdxz | ht.k_effxz/ht.C_eff | m²/s | Thermal diffusivity, xz component | Domains 2–4 | |
| ht.alphaTdyz | ht.k_effyz/ht.C_eff | m²/s | Thermal diffusivity, yz component | Domains 2–4 | |
| ht.alphaTdzz | ht.k_effzz/ht.C_eff | m²/s | Thermal diffusivity, zz component | Domains 2–4 | |
| ht.alphaTdMean | ht.kmean/ht.C_eff | m²/s | Mean thermal diffusivity | Domains 2–4 | |
| ht.Tradu | ht.Tu | K | Upside temperature | Domains 2–4 | |
| ht.Tradu | ht.Tu | K | Upside temperature | Boundaries 2–22 | |
| ht.Tradd | ht.Td | K | Downside temperature | Domains 2–4 | |
| ht.Tradd | ht.Td | K | Downside temperature | Boundaries 2–22 | |
| ht.dEi | material.dt(ht.rho*ht.Ei) | W/m³ | Total accumulated heat rate density | Domains 2–4 | |
| ht.dEi0 | material.dt(ht.rho*ht.Ei0) | W/m³ | Total accumulated energy rate density | Domains 2–4 | |
| ht.D_Hx | 0 | W/(m²·K) | Enthalpy diffusion coefficient, x component | Domains 2–4 | |
| ht.D_Hy | 0 | W/(m²·K) | | Domains 2–4 | |

| | | | | | |
|---|---|---|---|---|---|
| | | | Enthalpy diffusion coefficient, y component | | |
| ht.D_Hz | 0 | W/(m²·K) | Enthalpy diffusion coefficient, z component | Domains 2–4 | |
| ht.Q_H | ht.D_Hx*Tx+ht.D_Hy*Ty+ht.D_Hz*Tz | W/m³ | Enthalpy diffusion flux | Domains 2–4 | |
| ht.rholCplUlx | 0 | W/(m²·K) | Heat transfer coefficient, liquid water transport, x component | Domains 2–4 | |
| ht.rholCplUly | 0 | W/(m²·K) | Heat transfer coefficient, liquid water transport, y component | Domains 2–4 | |
| ht.rholCplUlz | 0 | W/(m²·K) | Heat transfer coefficient, liquid water transport, z component | Domains 2–4 | |
| ht.timeDerivative | material.dt(T) | K/s | Temperature, first time derivative | Domains 2–4 | |
| ht.gamma | 1 | 1 | Ratio of specific heats | Domains 2–4 | |
| ht.Trho | ht.Tref | K | Temperature for density evaluation | Domains 2–4 | |
| ht.dfltopaque | 1 | 1 | Default opacity | Domains 2–4 | |
| ht.helem | h_spatial | m | Element size | Domains 2–4 | |
| ht.solid1.pref | 1[atm] | Pa | Reference pressure level | Domains 2–4 | |
| ht.solid1.dEiInt | ht.solid1.intDom((ht.dEi-ht.Qm*ht.Ei)*ht.varIntSpa) | W | Total accumulated heat rate | Global | |
| ht.solid1.dEi0Int | ht.solid1.intDom((ht.dEi0-ht.Qm*ht.H)*ht.varIntSpa) | W | Total accumulated energy rate | Global | |
| ht.solid1.ntfluxInt | ht.solid1.intExtBnd(ht.ntflux*ht.varIntSpa)+ht.solid1.intExtBndUp(ht.ntflux_u*up(ht.varIntSpa))+ht.solid1.intExtBndDown(ht.ntflux_d*down(ht.varIntSpa))+ht.solid1.intIntBnd(ht.ncflux_u*up(ht.varIntSpa)+ht.ncflux_d*down(ht.varIntSpa)) | W | Total net heat rate | Global | |
| ht.solid1.ntefluxInt | ht.solid1.intExtBnd(ht.nteflux*ht.varIntSpa)+ht.solid1.intExtBndUp(ht.nteflux_u*up(ht.varIntSpa))+ht.solid1.intExtBndDown(ht.nteflux_d*down(ht.varIntSpa))+ht.solid1.intIntBnd(ht.nthflux_u*up(ht.varIntSpa)+ht.nthflux_d*down(ht.varIntSpa)) | W | Total net energy rate | Global | |
| ht.solid1.QInt | ht.solid1.intDom(ht.Qtot*ht.varIntSpa)+ht.solid1.intIntLine(ht.Qltot*ht.varIntSpa)+ht.intLine(subst(ht.Qlrtot,ht.id,isdefined(ht.solid1.id))*ht.varIntSpa)+ht.solid1.intIntPnt(ht.Qptot)+ht.intPnt(subst(ht.Qprtot,ht.id,isdefined(ht.solid1.id)))-ht.solid1.intIntBnd(ht.ndflux_u*up(ht.varIntSpa)+ht.ndflux_d*down(ht.varIntSpa)) | W | Total heat source | Global | |
| ht.solid1.WstrInt | ht.solid1.intDom(ht.Wstr*ht.varIntSpa) | W | Total stress power | Global | |
| ht.solid1.WInt | ht.solid1.intDom(ht.Wtot*ht.varIntSpa)+ht.solid1.intBndUp(ht.WBndTot_u*up(ht.varIntSpa))+ht.solid1.intBndDown(ht.WBndTot_d*down(ht.varIntSpa)) | W | Total work source | Global | |

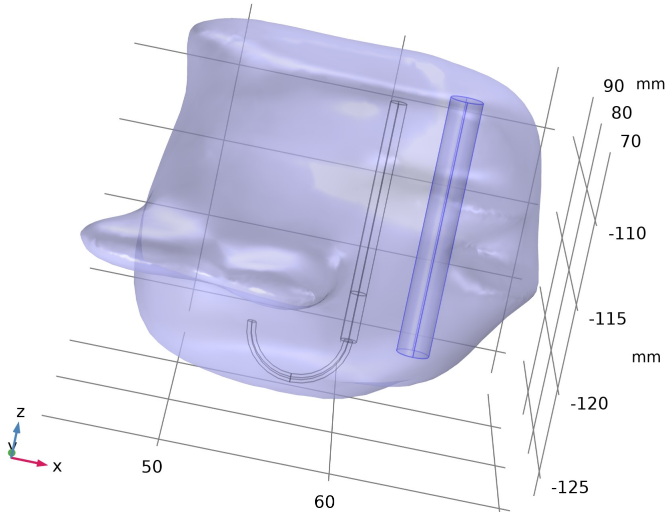
**Shape functions**

| Name | Shape function | Unit | Description | Shape frame | Selection |
|---|---|---|---|---|---|
| T | Lagrange (Quadratic) | K | Temperature | Spatial | Domains 2–4 |

**Weak Expressions**

| Weak expression | Integration order | Integration frame | Selection |
|---|---|---|---|
| ht.streamline | 4 | Spatial | Domains 2–4 |
| (ht.dfluxx*test(Tx)+ht.dfluxy*test(Ty)+ht.dfluxz*test(Tz))*ht.d | 4 | Spatial | Domains 2–4 |
| -ht.C_eff*ht.timeDerivative*test(T)*ht.d | 4 | Spatial | Domains 2–4 |
| -ht.C_eff*(ht.ux*Tx+ht.uy*Ty+ht.uz*Tz)*test(T)*ht.d | 4 | Spatial | Domains 2–4 |

**2.5.7. Temperature 1**




*Temperature 1*

Selection

| | |
|---|---|
| Geometric entity level | Boundary |
| Name | Tissue and Trocar, Exterior Boundaries |
| Selection | Named adj1: Geometry geom1: Input dimension 3 Output [EXTERIOR]: Domains 1–4; Dimension 2: Boundaries 1, 23–28 |

Equations

$$T = T_0$$

**Temperature**

Settings

| Description | Value |
|---|---|
| Temperature | User defined |
| Temperature | T_b |

**Variables**

| Name | Expression | Unit | Description | Selection | Details |
|---|---|---|---|---|---|
| ht.T0 | T_b | K | Temperature | Boundaries 1, 23–28 | + operation |
| ht.temp1.ntfluxInt | ht.temp1.intExtBnd(ht.ntflux*ht.varIntSpa)+ht.temp1.intIntBnd(ht.ncflux_u*up(ht.varIntSpa) +ht.ncflux_d*down(ht.varIntSpa)) | W | Total net heat rate | Global | |
| ht.temp1.ntefluxInt | ht.temp1.intExtBnd(ht.nteflux*ht.varIntSpa)+ht.temp1.intIntBnd(ht.nthflux_u*up(ht.varIntSpa) +ht.nthflux_d*down(ht.varIntSpa)) | W | Total net energy rate | Global | |
| ht.temp1.ntfluxInt_u | ht.temp1.intIntBnd(ht.ntflux_u*up(ht.varIntSpa)) | W | Total net heat rate, upside | Global | |
| ht.temp1.ntefluxInt_u | ht.temp1.intIntBnd(ht.nteflux_u*up(ht.varIntSpa)) | W | Total net energy rate, upside | Global | |
| ht.temp1.ntfluxInt_d | ht.temp1.intIntBnd(ht.ntflux_d*down(ht.varIntSpa)) | W | Total net heat rate, downside | Global | |
| ht.temp1.ntefluxInt_d | ht.temp1.intIntBnd(ht.nteflux_d*down(ht.varIntSpa)) | W | Total net energy rate, downside | Global | |
| ht.temp1.Tave | nojac(ht.temp1.intBnd(ht.varIntSpa*ht.rho*ht.Cp*T*max(abs (ht.ux*ht.nxmesh+ht.uy*ht.nymesh+ht.uz*ht.nzmesh),eps)))/nojac(ht.temp1.intBnd (ht.varIntSpa*ht.rho*ht.Cp*max(abs(ht.ux*ht.nxmesh+ht.uy*ht.nymesh+ht.uz*ht.nzmesh),eps))) | K | Weighted average temperature | Global | |

**Constraints**

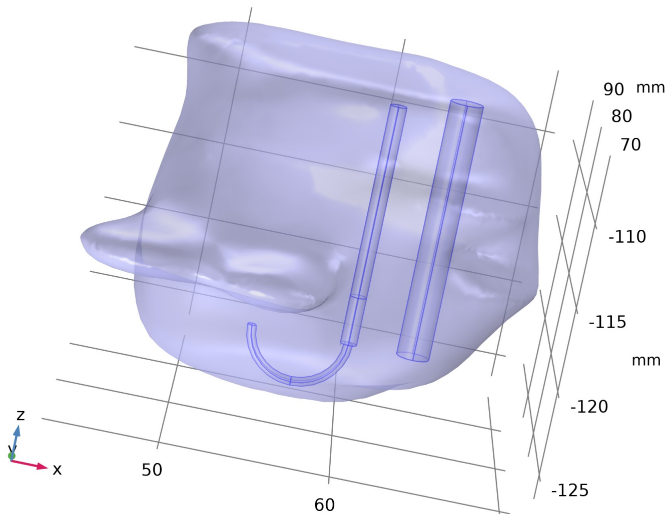
| Constraint | Constraint force | Shape function | Selection | Details |
|---|---|---|---|---|
| ht.T0-ht.Tvar | test(ht.T0-ht.Tvar) | Lagrange (Quadratic) | Boundaries 1, 23–28 | Elemental |

## 2.6. Multiphysics

### 2.6.1. Electromagnetic Heating 1

Used products

COMSOL Multiphysics




*Electromagnetic Heating 1*

Selection

| | |
|---|---|
| Geometric entity level | Domain |
| Selection | Geometry geom1: Dimension 3: All domains |

Equations

$$\rho C_p \frac{\partial T}{\partial t} + \rho C_p \mathbf{u} \cdot \nabla T = \nabla \cdot (k \nabla T) + Q_e$$

**Coupled Interfaces**

Settings

| Description | Value |
|---|---|
| Electromagnetic | Electric Currents (ec) |
| Heat transfer | Bioheat Transfer (ht) |

**Variables**

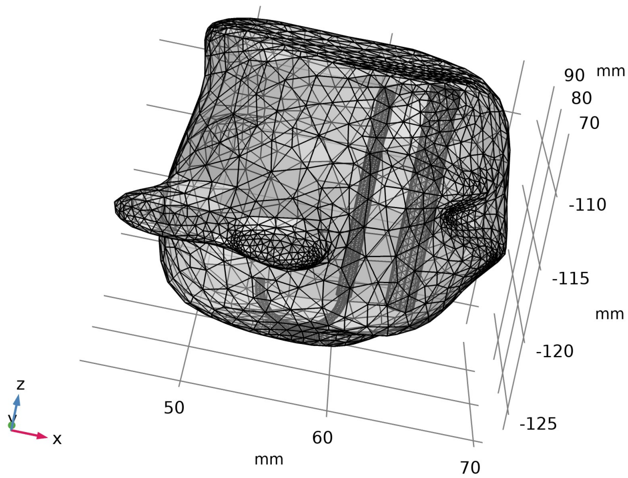
| Name | Expression | Unit | Description | Selection | Details |
|---|---|---|---|---|---|
| ht.ndflux_u | -0.5*emh1.Qb | W/m² | Internal normal conductive heat flux, upside | Boundaries 2–28 | + operation |
| ht.ndflux_d | -0.5*emh1.Qb | W/m² | Internal normal conductive heat flux, downside | Boundaries 2–28 | + operation |
| ht.Q | emh1.Q | W/m³ | Heat source | Domains 1–4 | + operation |
| ht.Qtot | emh1.Q | W/m³ | Total heat source | Domains 1–4 | + operation |
| ht.Qbtot | emh1.Qb | W/m² | Total boundary heat source | Boundaries 1–28 | + operation |
| ht.ntflux_contrib | emh1.Qb | W/m² | Boundary sources and fluxes contribution | Boundaries 2–28 | + operation |
| emh1.Q | ec.Qh | W/m³ | Heat source | Domains 1–4 | |
| emh1.T | T | K | Temperature | Global | |
| emh1.Qb | ec.Qsh | W/m² | Boundary heat source | Global | |
| emh1.varInt | 1 | | Intermediate variable | Boundaries 1–28 | Meta |
| emh1.ntfluxInt | emh1.intExtBnd(ht.ntflux*ht.varIntSpa)+emh1.intIntBnd(ht.ncflux_u*up(ht.varIntSpa) +ht.ncflux_d*down(ht.varIntSpa)) | W | Total net heat rate | Global | |
| emh1.ntefluxInt | emh1.intExtBnd(ht.nteflux*ht.varIntSpa)+emh1.intIntBnd(ht.nthflux_u*up(ht.varIntSpa) +ht.nthflux_d*down(ht.varIntSpa)) | W | Total net energy rate | Global | |
| emh1.ntfluxInt_u | emh1.intIntBnd(ht.ntflux_u*up(ht.varIntSpa)) | W | Total net heat rate, upside | Global | |
| | | | | | |

| | | | | | | |
|---|---|---|---|---|---|---|
| emh1.ntefluxInt_u | emh1.intIntBnd(ht.nteflux_u*up(ht.varIntSpa)) | W | Total net energy rate, upside | Global | |
| emh1.ntfluxInt_d | emh1.intIntBnd(ht.ntflux_d*down(ht.varIntSpa)) | W | Total net heat rate, downside | Global | |
| emh1.ntefluxInt_d | emh1.intIntBnd(ht.nteflux_d*down(ht.varIntSpa)) | W | Total net energy rate, downside | Global | |
| emh1.Tave | nojac(emh1.intBnd(ht.varIntSpa*ht.rho*ht.Cp*T*max(abs (ht.ux*ht.nxmesh+ht.uy*ht.nymesh+ht.uz*ht.nzmesh),eps)))/nojac(emh1.intBnd (ht.varIntSpa*ht.rho*ht.Cp*max(abs(ht.ux*ht.nxmesh+ht.uy*ht.nymesh+ht.uz*ht.nzmesh),eps))) | K | Weighted average temperature | Global | |
| emh1.Tvar | 0.5*(ht.Tu+ht.Td) | K | Temperature | Boundaries 1–28 | |

**Weak Expressions**

| Weak expression | Integration order | Integration frame | Selection |
|---|---|---|---|
| emh1.Q*test(T)*ht.d | 4 | Spatial | Domains 1–4 |
| emh1.Qb*test(emh1.Tvar) | 4 | Spatial | Boundaries 1–28 |

### 2.7. Mesh 1




*Mesh 1*

Mesh statistics

| Description | Value |
|---|---|
| Minimum element quality | 0.1908 |
| Average element quality | 0.695 |
| Tetrahedron | 58321 |
| Triangle | 5738 |
| Edge element | 538 |
| Vertex element | 32 |

#### 2.7.1. Size (size)

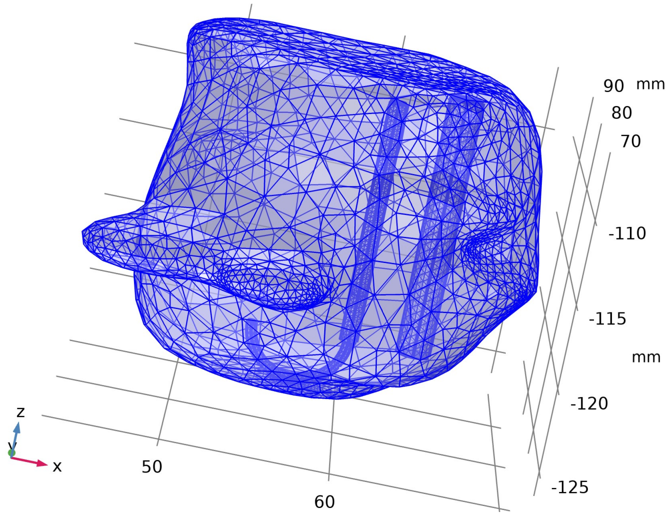
Settings

| Description | Value |
|---|---|
| Maximum element size | 2.44 |
| Minimum element size | 0.304 |
| Curvature factor | 0.5 |
| Resolution of narrow regions | 0.6 |
| Maximum element growth rate | 1.45 |
| Predefined size | Fine |

#### 2.7.2. Free Tetrahedral 1 (ftet1)

Selection

| | |
|---|---|
| Geometric entity level | Domain |
| Selection | Remaining |




*Free Tetrahedral 1*

Settings

| Description | Value |
|---|---|
| Avoid inverted curved elements | On |

## 3. Study 1

Computation information

| | |
|---|---|
| Computation time | 1 min 25 s |

### 3.1. Time Dependent

| Times | Unit |
|---|---|
| range(0,a_time/4,a_time) | min |

Study settings

| Description | Value |
|---|---|
| Include geometric nonlinearity | Off |

Study settings

| Description | Value |
|---|---|
| Time unit | min |
| Output times | {0, 2.5, 5, 7.5, 10} |

Physics and variables selection

| Physics interface | Discretization |
|---|---|
| Electric Currents (ec) | physics |
| Bioheat Transfer (ht) | physics |

Mesh selection

| Geometry | Mesh |
|---|---|
| Geometry 1 (geom1) | mesh1 |

### 3.2. Solver Configurations

#### 3.2.1. Solution 1

##### Compile Equations: Time Dependent (st1)

Study and step

| Description | Value |
|---|---|
| Use study | Study 1 |
| Use study step | Time Dependent |

Log

<---- Compile Equations: Time Dependent in Study 1/Solution 1 (sol1) -----------
Started at Apr 5, 2021 8:27:13 PM.
Geometry shape function: Linear Lagrange
Running on Intel64 Family 6 Model 142 Stepping 9, GenuineIntel.
Using 1 socket with 2 cores in total on DESKTOP-Q35TI0P.
Available memory: 12.17 GB.
Time: 5 s.

Physical memory: 1.6 GB
Virtual memory: 1.7 GB
Ended at Apr 5, 2021 8:27:18 PM.
----- Compile Equations: Time Dependent in Study 1/Solution 1 (sol1) ---------->

**Dependent Variables 1 (v1)**

General

| Description | Value |
|---|---|
| Defined by study step | Time Dependent |

Residual scaling

| Description | Value |
|---|---|
| Method | Manual |

Initial value calculation constants

| Constant name | Initial value source |
|---|---|
| t | range(0,a_time/4,a_time) |
| timestep | 0.01[min] |

Log
<---- Dependent Variables 1 in Study 1/Solution 1 (sol1) -----------------------
Started at Apr 5, 2021 8:27:18 PM.
Solution time: 0 s.
Physical memory: 1.6 GB
Virtual memory: 1.7 GB
Ended at Apr 5, 2021 8:27:18 PM.
----- Dependent Variables 1 in Study 1/Solution 1 (sol1) ---------------------->

**Degree of tissue injury (comp1.ht.alpha) (comp1_ht_alpha)**

General

| Description | Value |
|---|---|
| Field components | comp1.ht.alpha |

**Temperature (comp1.T) (comp1_T)**

General

| Description | Value |
|---|---|
| Field components | comp1.T |
| Internal variables | {comp1.uflux.T, comp1.dflux.T, comp1.ht.dt2Inv_T} |

**Electric potential (comp1.V) (comp1_V)**

General

| Description | Value |
|---|---|
| Field components | comp1.V |

**Time-Dependent Solver 1 (t1)**

General

| Description | Value |
|---|---|
| Defined by study step | Time Dependent |
| Time unit | min |
| Output times | {0, 2.5, 5, 7.5, 10} |

Field tolerance method

| Field | Value |
|---|---|
| Degree of tissue injury (comp1.ht.alpha) | Scaled |
| Temperature (comp1.T) | Use_global |
| Electric potential (comp1.V) | Use_global |

Field tolerance value method

| Field | Value |
|---|---|
| Degree of tissue injury (comp1.ht.alpha) | Manual |
| Temperature (comp1.T) | Factor |
| Electric potential (comp1.V) | Factor |

Field tolerance

| Field | Value |
|---|---|
| Degree of tissue injury (comp1.ht.alpha) | 1 |
| Temperature (comp1.T) | 1e-3 |
| Electric potential (comp1.V) | 1e-3 |

Time stepping

| Description | Value |
|---|---|
| | |

| | |
|---|---|
| Maximum BDF order | 2 |
| Error estimation | Exclude algebraic |

Log

```
<---- Time-Dependent Solver 1 in Study 1/Solution 1 (sol1) --------------------
Started at Apr 5, 2021 8:27:18 PM.
Time-dependent solver (BDF)
Number of degrees of freedom solved for: 137563 (plus 71864 internal DOFs).
Symmetric matrices found.
Scales for dependent variables:
Electric potential (comp1.V): 1
Orthonormal null-space function used.
Symmetric matrices found.
Scales for dependent variables:
Temperature (comp1.T): 3.1e+02
Orthonormal null-space function used.
Symmetric matrices found.
Scales for dependent variables:
Degree of tissue injury (comp1.ht.alpha): 1
Orthonormal null-space function used.
Symmetric matrices found.
Step    Time   Stepsize    Res  Jac  Sol Order Tfail NLfail LinIt  LinErr  LinRes
  0       0        - out   12    9   12           0
            Group #1:      4    3    4                 20   0.0084  9.1e-06
            Group #2:      4    3    4                     7.5e-16  4.6e-16
            Group #3:      4    3    4                  4  1.6e-13    4e-16
  1  0.045901   0.045901     18   12   18    1    0    0
            Group #1:      6    4    6                 26   0.0025        -
            Group #2:      6    4    6                     8.4e-16  5.2e-16
            Group #3:      6    4    6                  6  6.2e-17        -
  2  0.091803   0.045901     21   14   21    1    0    0
            Group #1:      7    4    7                 31   0.0087        -
            Group #2:      7    5    7                       1e-15  5.2e-16
            Group #3:      7    5    7                  7  3.9e-13        -
  3   0.18361   0.091803     24   16   24    2    0    0
            Group #1:      8    4    8                 36   0.0087        -
            Group #2:      8    6    8                     1.3e-15  5.2e-16
            Group #3:      8    6    8                  8  2.4e-13        -
  4   0.27541   0.091803     27   18   27    2    0    0
            Group #1:      9    4    9                 41   0.0088        -
            Group #2:      9    7    9                     1.1e-15    5e-16
            Group #3:      9    7    9                  9    3e-13        -
  5   0.45901    0.18361     30   21   30    1    0    0
            Group #1:     10    5   10                 46   0.0088        -
            Group #2:     10    8   10                     1.9e-15  1.8e-15
            Group #3:     10    8   10                 10  5.5e-13        -
  6   0.82623    0.36721     33   24   33    1    0    0
            Group #1:     11    6   11                 51   0.0089        -
            Group #2:     11    9   11                     4.5e-15  4.5e-15
            Group #3:     11    9   11                 11  3.3e-13        -
  7    1.5607    0.73442     39   27   39    1    0    0
            Group #1:     13    7   13                 60   0.0054        -
            Group #2:     13   10   13                     8.7e-15    7e-15
            Group #3:     13   10   13                 13  1.3e-16        -
  8    2.2951    0.73442     42   29   42    1    0    0
            Group #1:     14    7   14                 66   0.0011        -
            Group #2:     14   11   14                     4.6e-15  1.8e-14
            Group #3:     14   11   14                 14  6.2e-13        -
  9    3.7639     1.4688     48   32   48    1    0    0
            Group #1:     16    8   16                 76   0.0022        -
            Group #2:     16   12   16                       4e-15  2.6e-14
            Group #3:     16   12   16                 16  2.5e-17        -
 10    5.2328     1.4688     51   34   51    1    0    0
            Group #1:     17    8   17                 81   0.0088        -
            Group #2:     17   13   17                     8.7e-15  2.6e-14
            Group #3:     17   13   17                 17  1.7e-13        -
 11    8.1705     2.9377     57   37   57    1    0    0
            Group #1:     19    9   19                 91   0.0021        -
            Group #2:     19   14   19                     3.1e-14    4e-14
            Group #3:     19   14   19                 19  9.7e-18        -
 12    11.108     2.9377     60   39   60    1    0    0
            Group #1:     20    9   20                 97   0.0014        -
            Group #2:     20   15   20                     3.5e-15  3.8e-14
            Group #3:     20   15   20                 20  7.4e-14        -
 13    16.984     5.8754     66   42   66    1    0    0
            Group #1:     22   10   22                108   0.0031        -
            Group #2:     22   16   22                     3.7e-14  5.4e-14
            Group #3:     22   16   22                 22  7.9e-18        -
 14    22.859     5.8754     69   44   69    1    0    0
            Group #1:     23   10   23                113   0.0072        -
            Group #2:     23   17   23                     2.5e-14  4.8e-14
            Group #3:     23   17   23                 23  6.8e-14        -
 15     34.61     11.751     75   47   75    1    0    0
```

```
          Group #1:    25   11   25               123   0.0027      -
          Group #2:    25   18   25                  1.4e-14 6.7e-14
          Group #3:    25   18   25                25  5.2e-18      -
 16    46.36     11.751     78   49   78    1    0     0
          Group #1:    26   11   26               128   0.0071      -
          Group #2:    26   19   26                  3.9e-14 5.3e-14
          Group #3:    26   19   26                26  1.6e-13      -
 17   69.862     23.502     84   52   84    1    0     0
          Group #1:    28   12   28               138   0.0031      -
          Group #2:    28   20   28                  3.9e-14   7e-14
          Group #3:    28   20   28                28  7.7e-18      -
 18   93.364     23.502     87   54   87    1    0     0
          Group #1:    29   12   29               143   0.0087      -
          Group #2:    29   21   29                  7.4e-14 5.5e-14
          Group #3:    29   21   29                29  6.5e-14      -
 19   140.37     47.003     93   57   93    1    0     0
          Group #1:    31   13   31               153   0.0024      -
          Group #2:    31   22   31                  1.3e-14 6.2e-14
          Group #3:    31   22   31                31  3.3e-18      -
 -      150          - out
 20   187.37     47.003     96   59   96    1    0     0
          Group #1:    32   13   32               158    0.005      -
          Group #2:    32   23   32                  6.9e-14 4.7e-14
          Group #3:    32   23   32                32  6.8e-14      -
 21   247.37        60     99   61   99    1    0     0
          Group #1:    33   13   33               163   0.0038      -
          Group #2:    33   24   33                  4.8e-14 4.4e-14
          Group #3:    33   24   33                33  1.6e-13      -
 -      300          - out
 22   307.37        60    102   63  102    1    0     0
          Group #1:    34   13   34               168   0.0065      -
          Group #2:    34   25   34                  2.5e-14 3.7e-14
          Group #3:    34   25   34                34  1.8e-13      -
 23   367.37        60    105   65  105    1    0     0
          Group #1:    35   13   35               173   0.0075      -
          Group #2:    35   26   35                    3e-15 2.7e-14
          Group #3:    35   26   35                35  3.1e-14      -
 24   427.37        60    108   67  108    1    0     0
          Group #1:    36   13   36               178   0.0071      -
          Group #2:    36   27   36                  1.6e-15 1.6e-14
          Group #3:    36   27   36                36  1.4e-13      -
 -      450          - out
 25   487.37        60    111   69  111    1    0     0
          Group #1:    37   13   37               183   0.0066      -
          Group #2:    37   28   37                  4.2e-14 2.6e-14
          Group #3:    37   28   37                37  1.1e-13      -
 26   547.37        60    114   71  114    1    0     0
          Group #1:    38   13   38               188   0.0074      -
          Group #2:    38   29   38                  5.1e-15   4e-14
          Group #3:    38   29   38                38  4.1e-14      -
 -      600          - out
 27   607.37        60    117   73  117    1    0     0
          Group #1:    39   13   39               193   0.0084      -
          Group #2:    39   30   39                  5.3e-14 3.1e-14
          Group #3:    39   30   39                39  1.4e-13      -
Time-stepping completed.
Solution time: 80 s. (1 minute, 20 seconds)
Physical memory: 1.97 GB
Virtual memory: 2.12 GB
Ended at Apr 5, 2021 8:28:38 PM.
----- Time-Dependent Solver 1 in Study 1/Solution 1 (sol1) -------------------->
```

###### Segregated 1 (se1)

General

| Description | Value |
|---|---|
| Stabilization and acceleration | Anderson acceleration |
| Dimension of iteration space | 5 |

###### Electric Currents (ss1)

General

| Description | Value |
|---|---|
| Variables | Electric potential (comp1.V) |
| Linear solver | [Iterative 1] |

###### Temperature (ss2)

General

| Description | Value |
|---|---|
| Variables | Temperature (comp1.T) |
| Linear solver | Direct |

Method and termination

| Description | Value |
|---|---|
| Damping factor | 0.8 |
| Jacobian update | Once per time step |

###### Segregated Step 3 (ss3)

General

| Description | Value |
|---|---|
| Variables | Degree of tissue injury (comp1.ht.alpha) |
| Linear solver | iterative |

Method and termination

| Description | Value |
|---|---|
| Jacobian update | Once per time step |

###### Lower Limit 1 (ll1)

Lower limit

| Description | Value |
|---|---|
| Lower limits (field variables) | comp1.T 0 |

###### Iterative 1 (i1)

General

| Description | Value |
|---|---|
| Solver | Conjugate gradients |

###### Multigrid 1 (mg1)

General

| Description | Value |
|---|---|
| Solver | Algebraic multigrid |

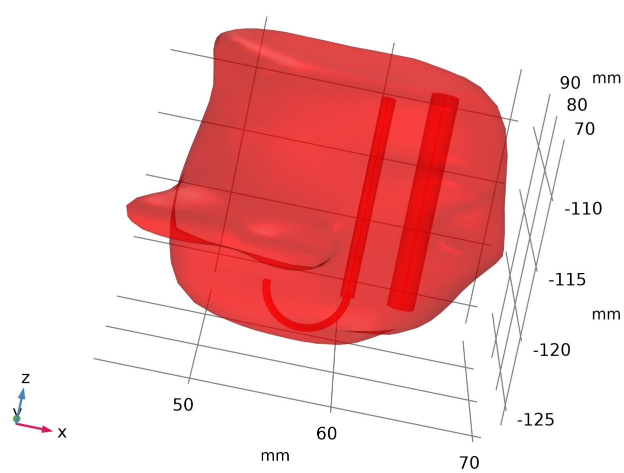
# 4. Results

## 4.1. Datasets

### 4.1.1. Study 1/Solution 1

Solution

| Description | Value |
|---|---|
| Solution | Solution 1 |
| Component | Component 1 (comp1) |

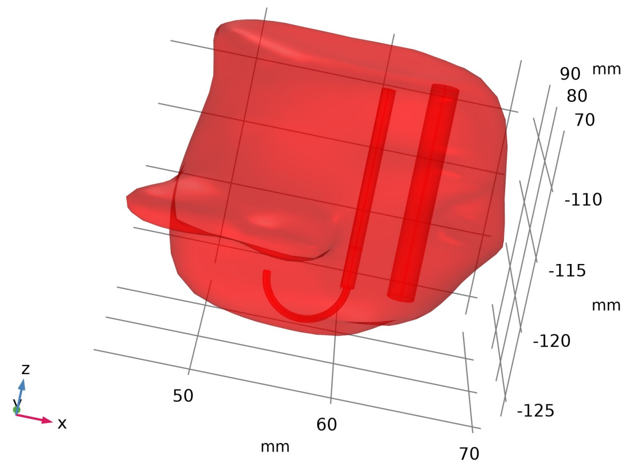


*Dataset: Study 1/Solution 1*

**4.1.2. Probe Solution 2**

Solution

| Description | Value |
|---|---|
| Solution | Solution 1 |
| Component | Component 1 (comp1) |
| Frame | Spatial (x, y, z) |




*Dataset: Probe Solution 2*

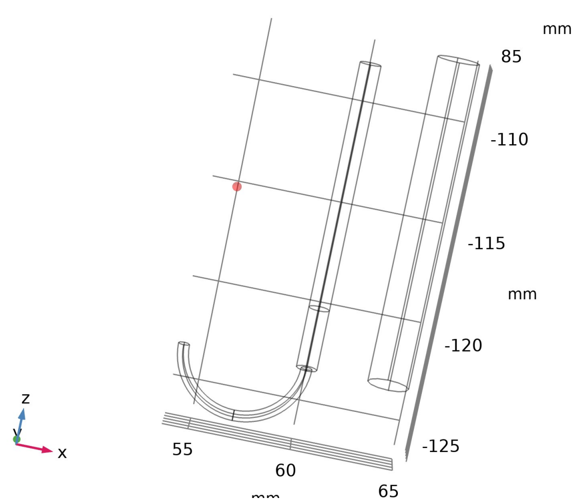
**4.1.3. Domain Point Probe 1**

Data

| Description | Value |
|---|---|
| Dataset | Probe Solution 2 |

Point data

| Description | Value |
|---|---|
| Entry method | Coordinates |
| x | 55 |
| y | 85 |
| z | -115 |




*Dataset: Domain Point Probe 1*

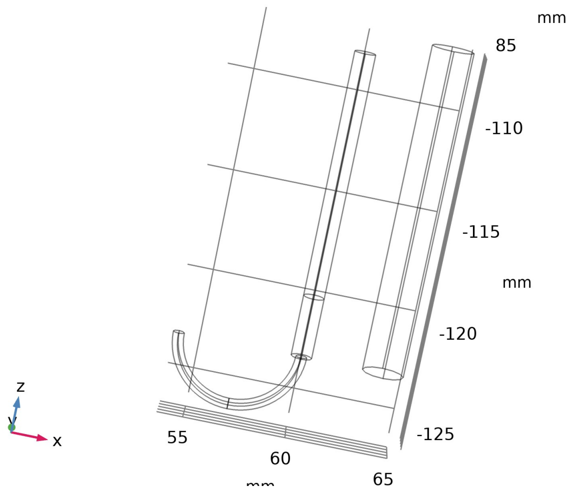
#### 4.1.4. Domain Point Probe 2

Data

| Description | Value |
|---|---|
| Dataset | Probe Solution 2 |

Point data

| Description | Value |
|---|---|
| Entry method | Coordinates |
| x | -12 |
| y | 0 |
| z | 65 |




*Dataset: Domain Point Probe 2*

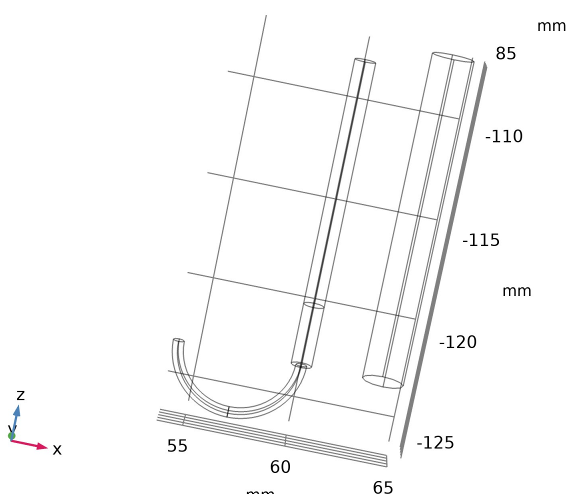
#### 4.1.5. Domain Point Probe 3

Data

| Description | Value |
|---|---|
| Dataset | Probe Solution 2 |

Point data

| Description | Value |
|---|---|
| Entry method | Coordinates |
| x | -20 |
| y | 0 |
| z | 65 |



*Dataset: Domain Point Probe 3*

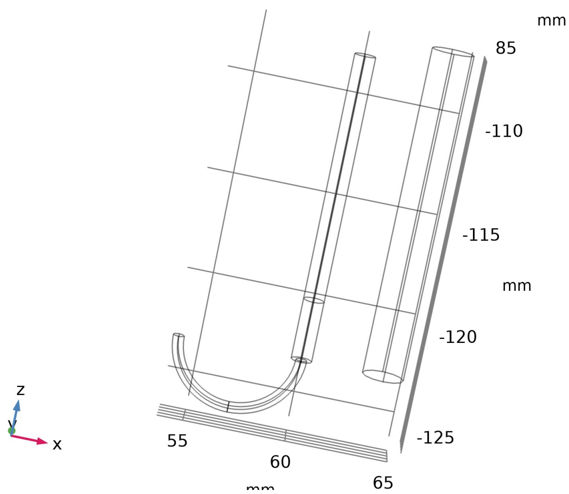
#### 4.1.6. Domain Point Probe 4

Data

| Description | Value |
|---|---|
| Dataset | Probe Solution 2 |

Point data

| Description | Value |
|---|---|
| Entry method | Coordinates |
| x | -0.2667 |
| y | 15.5 |
| z | 60 |




*Dataset: Domain Point Probe 4*

### 4.2. Derived Values

#### 4.2.1. Point Probe Expression 1

Output

| | |
|---|---|
| Evaluated in | Probe Table 1 |

Data

| Description | Value |
|---|---|
| Dataset | Domain Point Probe 1 |

Expressions

| Expression | Unit | Description |
|---|---|---|
| ht.theta_d | 1 | Fraction of damage |

#### 4.2.2. Point Probe Expression 1.3

Output

| | |
|---|---|
| Evaluated in | Probe Table 1 |

Data

| Description | Value |
|---|---|
| Dataset | Domain Point Probe 2 |

Expressions

| Expression | Unit | Description |
|---|---|---|
| ht.theta_d | 1 | Fraction of damage |

#### 4.2.3. Point Probe Expression 1.1

Output

| | |
|---|---|
| Evaluated in | Probe Table 1 |

Data

| Description | Value |
|---|---|
| Dataset | Domain Point Probe 3 |

Expressions

| Expression | Unit | Description |
|---|---|---|
| ht.theta_d | 1 | Fraction of damage |

#### 4.2.4. Point Probe Expression 4

Output

| | |
|---|---|
| Evaluated in | Probe Table 1 |

Data

| Description | Value |
|---|---|
| Dataset | Domain Point Probe 4 |

Expressions

| Expression | Unit | Description |
|---|---|---|
| T | degC | Temperature |

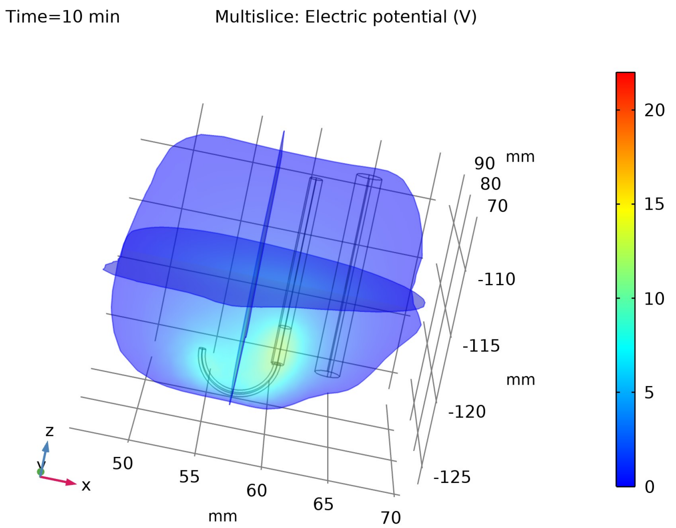
### 4.3. Tables

#### 4.3.1. Probe Table 1

| Time (min) | Fraction of damage (1), Point: (55, 85, -115) |
|---|---|
| 0.0000 | 3.5278E-7 |
| 7.6502E-4 | 1.3850E-5 |
| 0.0015300 | 2.7350E-5 |
| 0.0030601 | 5.4356E-5 |
| 0.0045901 | 8.1371E-5 |
| 0.0076502 | 1.3545E-4 |
| 0.013770 | 2.4378E-4 |
| 0.026011 | 4.6107E-4 |
| 0.038251 | 6.7901E-4 |
| 0.062732 | 0.0011175 |
| 0.087213 | 0.0015587 |
| 0.13617 | 0.0024534 |
| 0.18514 | 0.0033617 |
| 0.28306 | 0.0052564 |
| 0.38098 | 0.0072558 |
| 0.57683 | 0.011960 |
| 0.77267 | 0.017658 |
| 1.1644 | 0.034866 |
| 1.5561 | 0.059658 |
| 2.3394 | 0.13551 |
| 3.1228 | 0.22985 |
| 4.1228 | 0.35540 |
| 5.1228 | 0.46862 |
| 6.1228 | 0.56400 |
| 7.1228 | 0.64238 |
| 8.1228 | 0.70645 |
| 9.1228 | 0.75889 |
| 10.123 | 0.80188 |

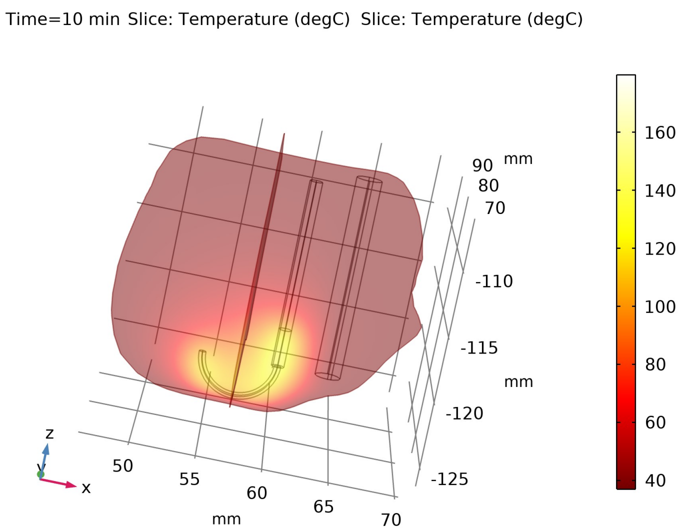
### 4.4. Plot Groups

#### 4.4.1. Electric Potential (ec)



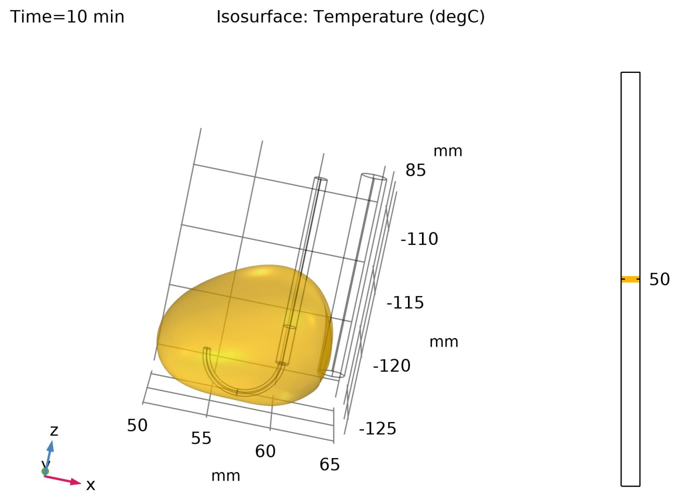

*Multislice: Electric potential (V)*

**4.4.2. Temperature (ht)**



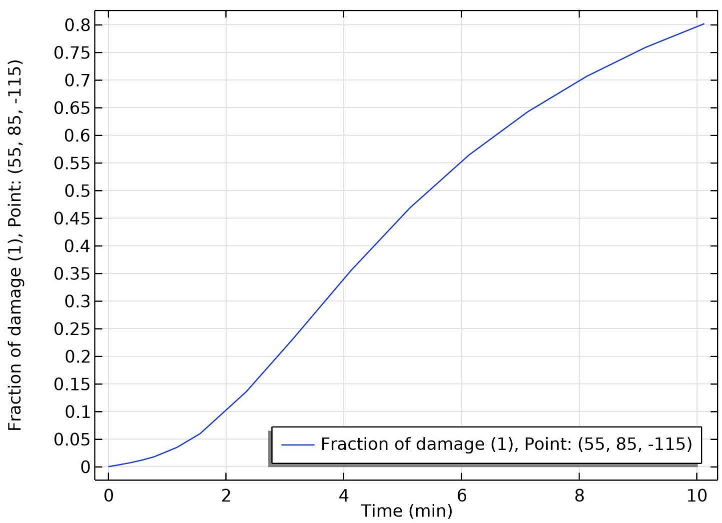

*Slice: Temperature (degC) Slice: Temperature (degC)*

**4.4.3. Isothermal Contours (ht)**

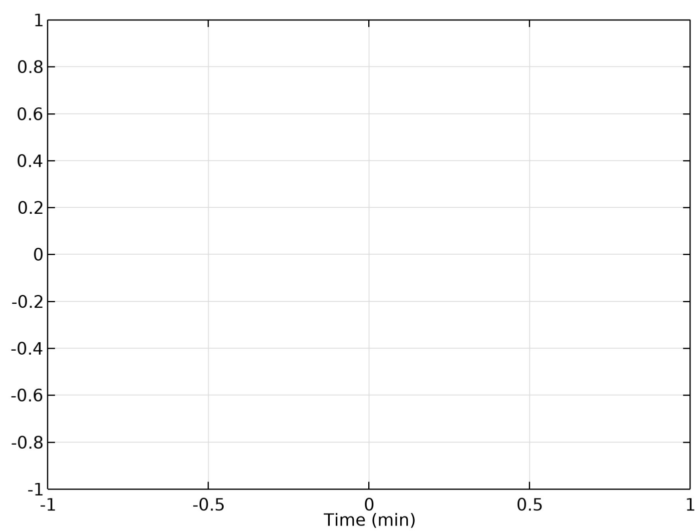



*Isosurface: Temperature (degC)*

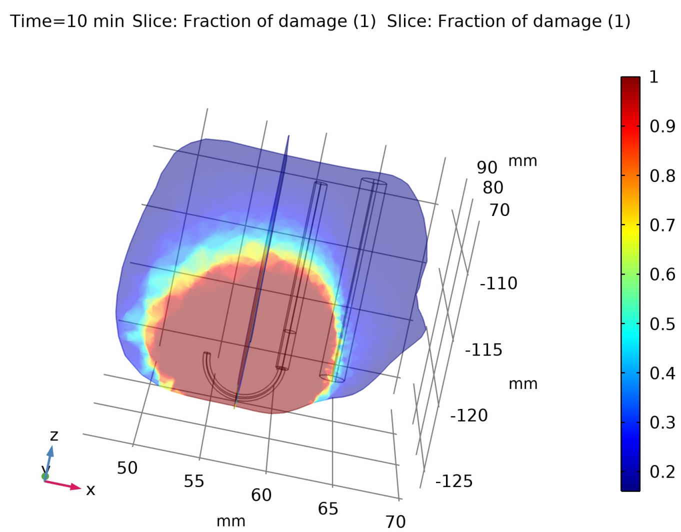
**4.4.4. Damaged Tissue, 1D**




**4.4.5. Temperature at One Electrode Tip**



#### 4.4.6. Damaged Tissue, 3D




*Slice: Fraction of damage (1) Slice: Fraction of damage (1)*